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Hypoxia-inducible factor induces cysteine dioxygenase and promotes cysteine homeostasis in *Caenorhabditis elegans*

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Abstract

Dedicated genetic pathways regulate cysteine homeostasis. For example, high levels of cysteine activate cysteine dioxygenase, a key enzyme in cysteine catabolism in most animal and many fungal species. The mechanism by which cysteine dioxygenase is regulated is largely unknown. In an unbiased genetic screen for mutations that activate cysteine dioxygenase (*cdo-1*) in the nematode *C. elegans*, we isolated loss-of-function mutations in *rhy-1* and *egl-9*, which encode proteins that negatively regulate the stability or activity of the oxygen-sensing hypoxia inducible transcription factor (*hif-1*). EGL-9 and HIF-1 are core members of the conserved eukaryotic hypoxia response. However, we demonstrate that the mechanism of HIF-1-mediated induction of *cdo-1* is largely independent of EGL-9 prolyl hydroxylase activity and the von Hippel-Lindau E3 ubiquitin ligase, the classical hypoxia signaling pathway components. We demonstrate that *C. elegans cdo-1* is transcriptionally activated by high levels of cysteine and *hif-1*. *hif-1*-dependent activation of *cdo-1* occurs downstream of an H_2S-sensing pathway that includes *rhy-1*, *cysl-1*, and *egl-9*. *cdo-1* transcription is primarily activated in the hypodermis where it is also sufficient to drive sulfur amino acid metabolism. Thus, the regulation of *cdo-1* by *hif-1* reveals a negative feedback loop that maintains cysteine homeostasis. High levels of cysteine stimulate the production of an H_2S signal. H_2S then acts through the *rhy-1/cysl-1/egl-9* signaling pathway to increase HIF-1-mediated transcription of *cdo-1*, promoting degradation of cysteine via CDO-1.

eLife assessment

The study presents **valuable** findings on how the hypoxia response pathway senses and responds to changes in the homeostasis of the amino acid cysteine and other sulfur-containing molecules. By providing a **compelling**, rigorous genetic analysis of the pathway, the study adds to a growing body of literature showing that prolyl hydroxylation is not the only mechanism by which the hypoxia response pathway can act. Although the paper does not reveal new biochemical insight into the mechanism, it opens up new areas of investigation that will be of interest to cell biologists and biomedical researchers studying the many pathologies involving hypoxia and/or cysteine metabolism.

Introduction

Cysteine is a sulfur-containing amino acid that mediates many oxidation/reduction reactions of proteins, is the redox center of the abundant antioxidant tripeptide glutathione which also serves as a major cysteine reserve, and is essential for iron-sulfur cluster assembly in the mitochondrion (1, 2). Cysteine residues in many proteins are in close proximity in the primary or folded protein sequence and are oxidized in the endoplasmic reticulum to form intra- and interprotein disulfide linkages, most commonly in secreted proteins which mediate intercellular signaling and defense (3–5). In many enzymes, the reactivity of the cysteine sulfur is key for the coordination of metals such as zinc or iron, which support protein structure and catalytic activity (6–8). Cysteine is also a key source of hydrogen sulfide (H_2S), a volatile signaling molecule (9). While cysteine has these essential functions, excess cysteine is also toxic. High levels of cysteine impair mitochondrial respiration by disrupting iron homeostasis (10), acts as a neural excitotoxin (11), and promotes the formation of toxic levels of hydrogen sulfide gas (9, 12, 13). Given this balance between essential and toxic, cysteine homeostasis is key for the health of cells and organisms.

CDO1-mediated oxidation is the primary pathway of cysteine catabolism when sulfur amino acid (methionine or cysteine) availability is normal or high (14). The dipeptide cystathionine is a key intermediate in this pathway. Cystathionine is catabolized by cystathionase (CTH-2 in *C. elegans*, CTH in mammals) producing cysteine and α-ketobutyrate. Cysteine is further oxidized using dissolved atmospheric dioxygen to cysteinesulfinate by cysteine dioxygenase (CDO-1 in *C. elegans*, CDO1 in mammals) (**Fig. 1A**) (15). The further oxidation of cysteinesulfinate downstream of CDO-1 generates highly toxic sulfites that are normally oxidized to more benign sulfate by sulfite oxidase (**Fig. 1A**).

As a critical player in cysteine homeostasis, CDO1 is a highly regulated enzyme: the activity and abundance of CDO1 increase dramatically in cells and animals fed excess cysteine and methionine (16). CDO1 levels are governed both transcriptionally and post-translationally via proteasomal degradation (17–21). However, the molecular players that sense high levels of cysteine and promote CDO1 activation remain incompletely defined.

Understanding these regulatory mechanisms is an important goal as CDO1 dysfunction is implicated in disease. CDO1 is a tumor suppressor whose activity is silenced in diverse cancers via promoter methylation, a potential biomarker of tumor grade and progression (22–24). Decreased CDO1 activity may support tumor cell growth by reducing reactive oxygen species and decreasing drug susceptibility (25–28).

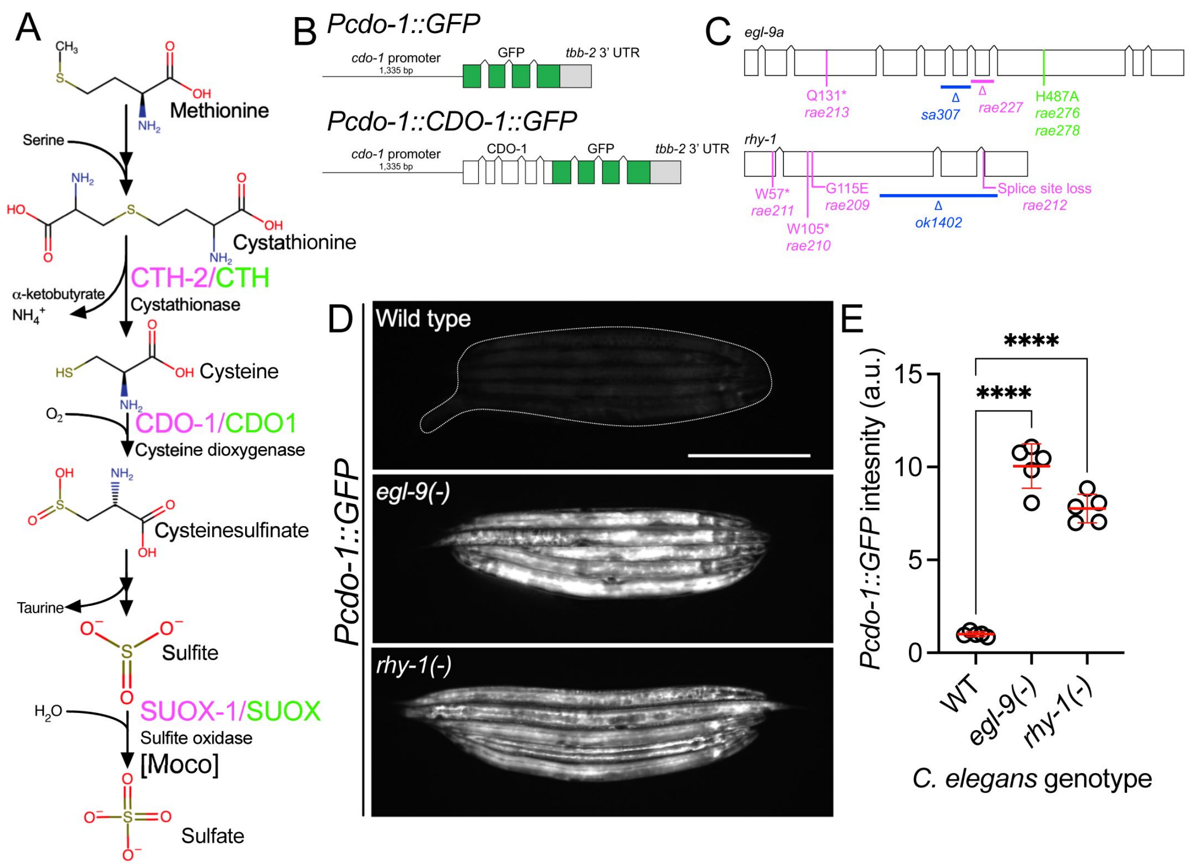

Figure 1

egl-9 and *rhy-1* inhibit *cdo-1* transcription.

A) Pathway for sulfur amino acid metabolism beginning with methionine. We highlight the roles of cystathionase (CTH-2/CTH), cysteine dioxygenase (CDO-1/CDO1), and the Moco-requiring sulfite oxidase enzyme (SUOX-1/SUOX). *C. elegans* enzymes (magenta) and their human homologs (green) are displayed. B) *Pcdo-1::GFP* promoter fusion (upper) and *Pcdo-1::CDO-1::GFP* C-terminal protein fusion (lower) transgenes used in this work are displayed. Boxes indicate exons, connecting lines indicate introns. The *cdo-1* promoter is shown as a straight line. C) *egl-9a* and *rhy-1* gene structures. Boxes indicate exons and connecting lines are introns. Colored annotations indicate mutations generated or used in our work. Magenta; chemically-induced mutations that activated *Pcdo-1::CDO-1::GFP* fusion protein. Blue; reference null alleles isolated independent of our work. Green; CRISPR/Cas9-generated mutation that inactivates the prolyl hydroxylase domain of EGL-9. D) Expression of *Pcdo-1::GFP* transgene is displayed for wild-type, *egl-9(sa307),* and *rhy-1(ok1402) C. elegans* animals at the L4 stage. Scale bar is 250μm. White dotted line outlines animals with basal GFP expression. For GFP imaging, exposure time was 100ms. E) Quantification of GFP expression displayed in **Fig. 1D**. Individual datapoints are shown (circles) as are the mean and standard deviation (red lines). *n* is 5 individuals per genotype. Data are normalized so that wild-type expression of *Pcdo-1::GFP* is 1 arbitrary unit (a.u.). ****, p<0.0001, ordinary one-way ANOVA with Dunnett's post hoc analysis.

Using the nematode *C. elegans*, we have previously shown that CDO-1 is a key player in the pathophysiology of two fatal inborn errors of metabolism; isolated sulfite oxidase deficiency (ISOD) and molybdenum cofactor deficiency (MoCD) (29, 30). Molybdenum cofactor (Moco) is an essential 520 Dalton prosthetic group synthesized from GTP by a conserved multistep biosynthetic pathway that is present in about 2/3 of bacterial genomes and nearly all eukaryotic genomes (31, 32). *C. elegans* can either retrieve Moco synthesized by the bacteria it consumes or can synthesize Moco *de novo* using its own Moco biosynthetic pathway (33). *C. elegans* strains carrying mutations in genes encoding Moco biosynthetic enzymes (*moc*) and feeding on wild-type *E. coli* develop normally. Yet, these same *moc*-mutant *C. elegans* when fed on Moco-deficient *E. coli* as their sole nutritional source are inviable, arresting development at an early larval stage. A saturated genetic selection for mutations that suppress this inviability identified multiple independent mutations in *cth-2* or *cdo-1*, genes which encode enzymes in the cysteine biosynthetic and degradation pathway. The toxic sulfites produced downstream of CDO-1 are normally oxidized to more benign sulfate by Moco-requiring sulfite oxidase (SUOX-1 in *C. elegans*, SUOX in mammals) an essential enzyme in *C. elegans* and humans (**Fig. 1A**) (30, 33). Therefore, loss of *cdo-1* or *cth-2* suppresses the lethality caused by both Moco and sulfite oxidase deficiencies in *C. elegans* by preventing the production of toxic sulfites (33–35). Thus, understanding the fundamental mechanisms that govern the levels and activity of CDO-1 is critical to generating new therapeutic hypotheses to treat these diseases.

To define genes that regulate *cdo-1* levels and activity, we performed an unbiased genetic screen in the nematode *C. elegans* to identify mutations that increase the expression or abundance of a *Pcdo-1::CDO-1::GFP* reporter transgene. We identified multiple independent loss-of-function mutations in two genes, *egl-9* and *rhy-1*, that dramatically increase expression of this P*cdo-1::CDO-1::GFP* transgene. These enzymes act in the hypoxia and H_2S-sensing pathway, and we demonstrate that the conserved hypoxia-inducible transcription factor (HIF-1) activates *cdo-1* transcription in this pathway (36–38). We further show that high levels of cysteine promote *cdo-1* transcription, and that *hif-1* and *cysl-1* (another component of the H_2S-sensing pathway) are required for viability under high cysteine conditions. We demonstrate that transcriptional activation of *cdo-1* via HIF-1 promotes CDO-1 activity and establish the *C. elegans* hypodermis as a key tissue of CDO-1 activation and function. Unexpectedly, we find that *cdo-1* regulation is governed by a HIF-1 pathway largely independent of EGL-9 prolyl hydroxylase activity and von Hippel-Lindau (VHL-1), the canonical O_2 -sensing pathway (39, 40). These data establish a new connection between the HIF-1/H_2S-sensing pathway and sulfur amino acid catabolism governed by CDO-1.

Results

egl-9 and *rhy-1* negatively regulate *cdo-1* transcription

To identify regulators of CDO-1 expression or activity, we engineered a transgene expressing a C-terminal green fluorescent protein (GFP) fusion to the full-length CDO-1 protein driven by the native *cdo-1* promoter (*Pcdo-1::CDO-1::GFP*, **Fig. 1B**). Transgenic animals were generated by integrating the *Pcdo-1::CDO-1::GFP* fusion protein into the *C. elegans* genome (41). The *Pcdo-1::CDO-1::GFP* fusion protein was functional and rescued a *cdo-1* loss of function mutation: the *Pcdo-1::CDO-1::GFP* fusion protein reverses the suppression of Moco-deficient lethality caused by *cdo-1* loss of function (**Fig. S1**). The reanimation of Moco-deficient lethality by the transgene depends on CDO-1 enzymatic activity because a transgene expressing an active-site mutant *Pcdo-1::CDO-1[C85Y]::GFP* does not rescue the *cdo-1* mutant phenotype (**Fig. S1**) (33, 42). Thus, the *Pcdo-1::CDO-1::GFP* transgenic fusion protein is functional, suggesting that its expression pattern reflects endogenous protein expression, localization, and levels.

We performed a mutagenesis screen to identify genes that control the expression or accumulation of CDO-1 protein. Specifically, we performed an EMS chemical mutagenesis of *C. elegans* and screened in the F2 generation, after newly induced random mutations were allowed to become homozygous, for mutations that caused increased GFP accumulation by the *Pcdo-1::CDO-1::GFP* reporter transgene (43). Using whole-genome sequencing, we determined that two independently isolated mutants carried distinct mutations in *egl-9* and four other independently isolated mutants had unique mutations in *rhy-1* (**Fig. 1C**). The presence of multiple independent alleles suggests these mutations in *egl-9* or *rhy-1* are causative for the increased *Pcdo-1::CDO-1::GFP* expression or accumulation observed. *egl-9* encodes the O_2 -sensing prolyl hydroxylase and orthologue of mammalian EglN1. *rhy-1* encodes the regulator of hypoxia inducible transcription factor, an enzyme with homology to membrane-bound O-acyltransferases (36, 39). The genetic screen produced nonsense alleles of both *egl-9* and *rhy-1* suggesting the increased *Pcdo-1::CDO-1::GFP* expression or accumulation is caused by loss of *egl-9* or *rhy-1* function (**Fig. 1C**). Inactivation of *egl-9* or *rhy-1* activates a transcriptional program mediated by the hypoxia inducible transcription factor, HIF-1 (36, 39). To determine if *egl-9* or *rhy-1* regulate *cdo-1* transcription, we engineered a separate reporter construct where only GFP (rather than the full length CDO-1::GFP fusion protein) is transcribed by the *cdo-1* promoter (*Pcdo-1::GFP,* **Fig. 1B**). This *Pcdo-1::GFP* transgene was introduced into strains with independently isolated *egl-9(sa307)* and *rhy-1(ok1402)* null reference alleles (36, 44). The *egl-9(sa307)* and *rhy-1(ok1402)* mutations dramatically induce expression of GFP driven by the *Pcdo-1::GFP* transcriptional reporter transgene (**Fig. 1D,E**). The activation of *cdo-1* transcription by independently isolated *egl-9* or *rhy-1* mutations demonstrates that the *egl-9* and *rhy-1* mutations isolated in our screen for the induction or accumulation of *Pcdo-1::CDO-1::GFP* are the causative genetic lesions. Furthermore, these data demonstrate that *egl-9* and *rhy-1* are necessary for the normal transcriptional repression of *cdo-1*.

HIF-1 directly activates *cdo-1* transcription downstream of the *rhy-1, cysl-1, egl-9* genetic pathway

rhy-1 and *egl-9* act in a pathway that regulates the abundance and activity of the HIF-1 transcription factor (36, 38). The activity of *rhy-1* is most upstream in the pathway and negatively regulates the activity of *cysl-1,* which encodes a cysteine synthase-like enzyme of probable algal origin (45). CYSL-1 directly binds to and inhibits EGL-9 in an H_2S-modulated manner (38). EGL-9 uses molecular oxygen as well as an α-ketoglutarate cofactor to directly inhibit HIF-1 via prolyl hydroxylation, which recruits the VHL-1 ubiquitin ligase to ubiquitinate HIF-1, targeting it for degradation by the proteasome **(Fig. 5A)** (46–48). Given that loss-of-function mutations in *rhy-1* or *egl-9* activate *cdo-1* transcription, we tested if *cdo-1* transcription is activated by HIF-1 as an output of this hypoxia/H_2S-sensing pathway. We performed epistasis studies using null mutations that inhibit the activity of *hif-1* (*cysl-1(ok762)* and *hif-1(ia4)*) or activate *hif-1* (*rhy-1(ok1402)* and *egl-9(sa307)*). The induction of *Pcdo-1::GFP* by *egl-9* inactivation was dependent upon the activity of *hif-1,* but not on the activity of *cysl-1* (**Fig. 2A,B**). In contrast, induction of *Pcdo-1::GFP* by *rhy-1* inactivation was dependent upon the activity of both *hif-1* and *cysl-1* (**Fig. 2A,C**). These results reveal a genetic pathway whereby *rhy-1*, *cysl-1*, and *egl-9* function in a negative-regulatory cascade to control the activity of HIF-1 which transcriptionally activates *cdo-1.* Our epistasis studies of *cdo-1* transcriptional regulation by HIF-1 align well with previous analyses of this genetic pathway in the context of transcription of *cysl-2* (a paralog of *cysl-1*) and the "O_2-ON response" (38).

To demonstrate that HIF-1 activates transcription of endogenous *cdo-1*, we explored published RNA-sequencing data of wild-type, *egl-9(-),* and *egl-9(-) hif-1(-)* mutant animals (49). *egl-9(-)* mutant *C. elegans* display an 8-fold increase in *cdo-1* mRNA compared to wild type. This induction was dependent on *hif-1;* a *hif-1(-)* mutation completely suppressed the induction of *cdo-1* mRNA caused by an *egl-9(-)* mutation (49). These RNA-seq data confirm our findings using the *Pcdo-1::GFP* transcriptional reporter that HIF-1 is a transcriptional activator of *cdo-1.* ChIP-seq data of

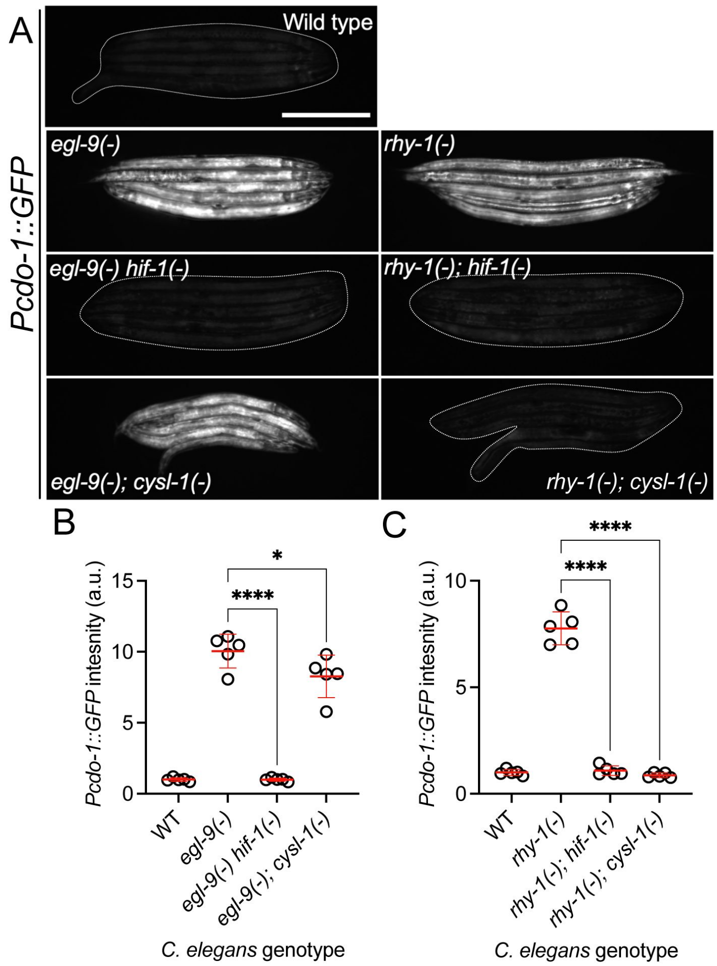

Figure 2

cdo-1 transcription is activated by HIF-1 downstream of RHY-1, CYSL-1, and EGL-9.

A) Expression of the *Pcdo-1::GFP* transgene is displayed for wild-type, *egl-9(sa307), egl-9(sa307) hif-1(ia4)* double mutant, *egl-9(sa307); cysl-1(ok762)* double mutant, *rhy-1(ok1402), rhy-1(ok1402); hif-1(ia4)* double mutant, and *rhy-1(ok1402); cysl-1(ok762)* double mutant *C. elegans* animals at the L4 stage of development. Scale bar is 250μm. White dotted line outlines animals with basal GFP expression. For GFP imaging, exposure time was 100ms. B,C) Quantification of the data displayed in **Fig. 2A**. Individual datapoints are shown (circles) as are the mean and standard deviation (red lines). *n* is 5 individuals per genotype. Data are normalized so that wild-type expression of *Pcdo-1::GFP* is 1 arbitrary unit (a.u.). *, p<0.05, ****, p<0.0001, ordinary one-way ANOVA with Dunnett's post hoc analysis. Note, wild-type, *egl-9(-)*, and *rhy-1(-)* images in panel A and quantification of *Pcdo-1::GFP* in panels B and C are identical to the data presented in **Fig. 1D,E**. They are re-displayed here to allow for clear comparisons to the double mutant strains of interest.

HIF-1 performed by the modERN project show that HIF-1 directly binds the *cdo-1* promoter (peak from −1,165 to −714 base pairs 5' to the *cdo-1* ATG start codon) (50, 51). This HIF-1 binding site contains 3 copies of the HIF-binding motif (5'-RCGTG-3') (52). Thus, *cdo-1* is a downstream effector of HIF-1 and is likely a direct transcriptional target of HIF-1.

High levels of cysteine promote *cdo-1* transcription and cause lethality in *cysl-1* and *hif-1* mutant animals

Mammalian CDO1 levels and activity are highly induced by dietary cysteine (14–16). To determine if this homeostatic response is conserved in *C. elegans*, we exposed transgenic *C. elegans* carrying the *Pcdo-1::GFP* transcriptional reporter to high supplemental cysteine. Like our *egl-9* and *rhy-1* loss-of-function mutations, high levels of cysteine promoted *cdo-1* transcription (**Fig. 3A-C**). Despite the significant 3.6-fold induction of *Pcdo-1::GFP* caused by 100μM supplemental cysteine, we note that this induction is not as dramatic as the induction caused by null mutations in *egl-9* or *rhy-1*. This perhaps reflects the animals' ability to buffer environmental cysteine which is bypassed by genetic intervention.

We hypothesized that cysteine might activate *cdo-1* transcription through the RHY-1/CYSL-1/EGL-9/HIF-1 pathway. In this pathway, *cysl-1* and *hif-1* act to promote *cdo-1* transcription. Thus, we sought to test whether *cysl-1* or *hif-1* were necessary for the induction of *Pcdo-1::GFP* by high levels of cysteine. However, this experiment was not possible as we observed 100% lethality in *cysl-1(-)* and *hif-1(-)* mutant animals exposed to 100μM supplemental cysteine, a cysteine concentration at which wild-type animals are healthy (**Fig. 3E**). Importantly, wild-type, *cysl-1(-)* and *hif-1(-)* animals were all healthy under control conditions without supplemental cysteine (**Fig. 3D**). While this phenotype limits our ability to establish the role of *cysl-1* and *hif-1* in the induction of *cdo-1* by high levels of cysteine, these data demonstrate that *cysl-1* and *hif-1* are necessary for survival under high cysteine conditions. Given the requirement of *hif-1* for survival in high levels of cysteine, we hypothesized that mutations in *egl-9* or *rhy-1* that activate *hif-1* might promote cysteine resistance. Indeed, we found that *egl-9(-)* and *rhy-1(-)* mutant *C. elegans* were partially viable when exposed to 1,000μM supplemental cysteine, a concentration that causes 100% lethality in wild-type animals (**Fig. 3F**). Thus, *egl-9* and *rhy-1* are negative regulators cysteine tolerance. Taken together, these data demonstrate a critical physiological role for the RHY-1/CYSL-1/EGL-9/HIF-1 pathway in promoting cysteine homeostasis.

To further test the interaction between high levels of cysteine and the RHY-1/CYSL-1/EGL-9/HIF-1 pathway, we exposed *egl-9(-); Pcdo-1::GFP* mutant animals to control or high levels of supplemental cysteine. We reasoned that if cysteine and *egl-9* loss of function promote *Pcdo-1::GFP* accumulation in the same pathway, then their effects should not be additive. Consistent with this hypothesis, we saw no difference in *Pcdo-1::GFP* expression in *egl-9(-)* mutant animals exposed to 0 or 100μM supplemental cysteine (**Fig. 3A,B**). We also tested the ability of supplemental cysteine to further activate *Pcdo-1::GFP* expression in *rhy-1(-)* mutant animals. In contrast to our results with *egl-9(-)*, we observed that high levels of cysteine caused a significant induction of *Pcdo-1::GFP* in the *rhy-1(-)* mutant background. These data suggest that cysteine acts in a pathway with *egl-9* but operates in parallel to the function of *rhy-1*.

Given the established role of CDO-1 in cysteine catabolism, we tested whether *cdo-1(-)* mutants were also sensitive to high levels of cysteine. *cdo-1(-)* mutant animals were not sensitive to high levels of supplemental cysteine compared to the wild type (**Fig. 3D-F**). Thus, *cdo-1* is not necessary for survival under high cysteine conditions. This suggests the existence of alternate pathways that promote cysteine homeostasis. Given the role of the RHY-1/CYSL-1/EGL-9/HIF-1 pathway in promoting cysteine homeostasis, we propose that HIF-1 activates pathways (in addition to *cdo-1*) that promote survival under high cysteine conditions.

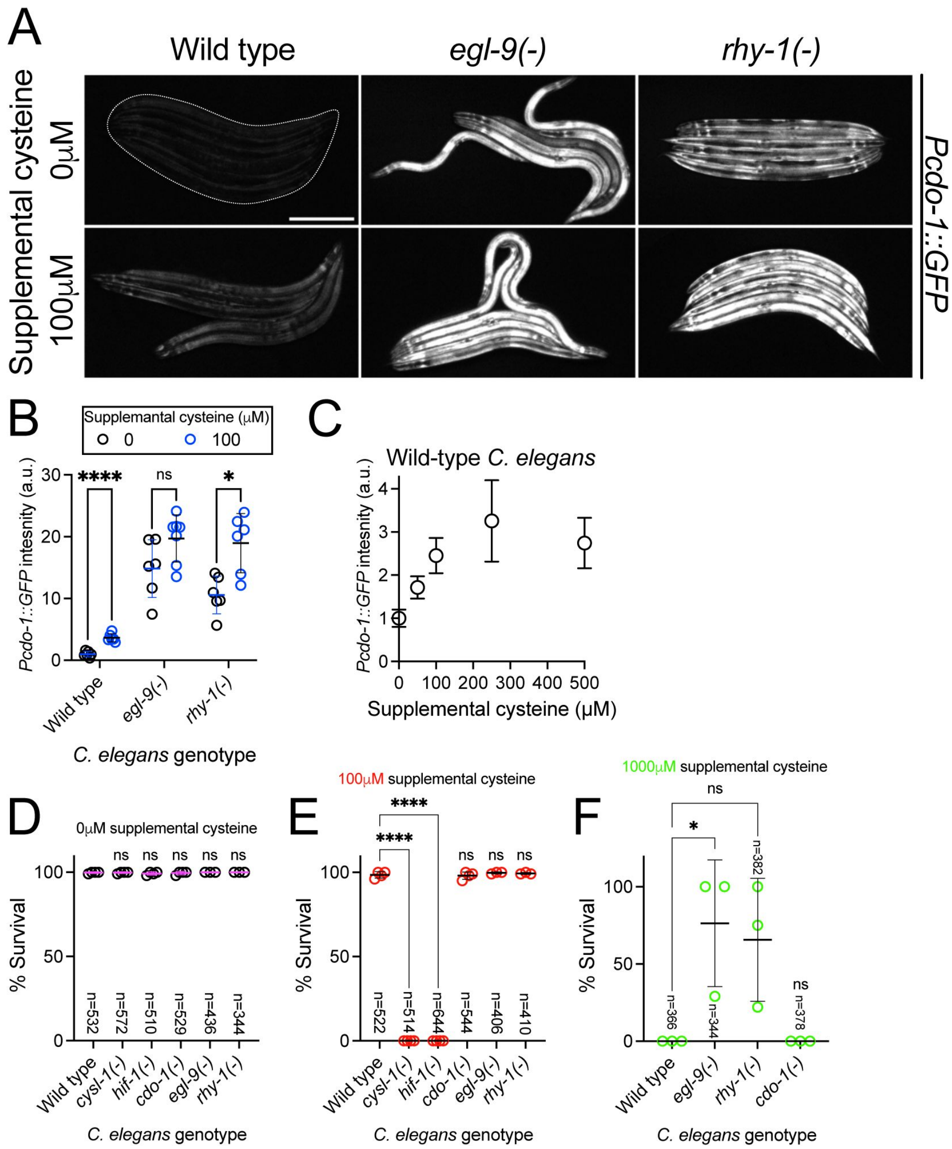

Figure 3

High levels of cysteine activate *cdo-1* transcription and are lethal to *hif-1* and *cysl-1* mutant animals.

A) Expression of the *Pcdo-1::GFP* transgene is displayed for wild-type, *egl-9(sa307),* and *rhy-1(ok1402)* young-adult *C. elegans* exposed to 0 or 100μM supplemental cysteine. Scale bar is 250μm. White dotted line exposure time was 100ms. Supplemental cysteine did not impact the mortality of the animals being imaged. B) Quantification of the data displayed in **Fig. 3A**. Individual datapoints are shown (circles) as are the mean and standard deviation (black lines). *n* is 6 or 7 individuals per genotype. Data are normalized so that expression of *Pcdo-1::GFP* in wild-type *C. elegans* exposed to 0μM supplemental cysteine is equal to 1 arbitrary unit (a.u.). *, p<0.05, ****, p<0.0001, multiple unpaired t test with Welch's correction. C) Quantification of the *Pcdo-1::GFP* expression is displayed for wild-type young-adult *C. elegans* exposed to 0, 50, 100, 250 or 500μM supplemental cysteine. Mean and standard deviation are displayed. n is 6 or 7 individuals per concentration of supplemental cysteine. D-F) The percentage of animals that survive overnight exposure to D) 0, E) 100, or F) 1,000μM supplemental cysteine. Individual datapoints (circles) represent biological replicates. 3 or 4 biological replicates were performed for each experiment and the total individuals scored amongst all replicates is displayed (n). *, p<0.05, ****, p<0.0001, ordinary one-way ANOVA with Dunnett's post hoc analysis. ns indicates no significant difference was identified.

Activated CDO-1 accumulates and is functional in the hypodermis

We sought to identify the site of action of CDO-1. To observe CDO-1 localization, we used CRISPR/Cas9 to insert the GFP open reading frame into the endogenous *cdo-1* locus, replacing the native *cdo-1* stop codon. This *cdo-1(rae273)* allele encodes a C-terminal tagged “CDO-1::GFP” fusion protein from the native *cdo-1* genomic locus (**Fig. 4A**). To determine if CDO-1::GFP was functional, we combined the CDO-1::GFP fusion protein with a null mutation in *moc-1*, a gene that is essential for *C. elegans* Moco biosynthesis (33). We then observed the growth of *moc-1(-)* CDO-1::GFP *C. elegans* on wild-type and Moco-deficient *E. coli.* The *moc-1(-)* mutant *C. elegans* expressing CDO-1::GFP from the native *cdo-1* locus arrest during larval development when fed Moco-deficient *E. coli*, but not when fed Moco-producing *E.coli* (**Fig. 4C**). This lethality is caused by the CDO-1-mediated production of sulfites which are only toxic when *C. elegans* is Moco deficient, and demonstrates that the CDO-1::GFP fusion protein is functional (33). These data suggest that the CDO-1::GFP expression we observe is physiologically relevant.

When wild-type animals expressing CDO-1::GFP were grown under standard culture conditions, we observed CDO-1::GFP expression in multiple tissues, including prominent expression in the hypodermis (**Fig. 4B**, **Fig. S2**). We tested if CDO-1::GFP fusion protein levels were affected by inactivation of *egl-9* or *rhy-1*. We generated *egl-9(-)*; CDO-1::GFP and *rhy-1(-)*; CDO-1::GFP animals, and assayed expression of CDO-1::GFP. We found that CDO-1::GFP fusion protein levels, encoded by *cdo-1(rae273)*, were increased by *egl-9(-)* or *rhy-1(-)* mutations (**Fig. 4B**, **Fig. S2**). This is consistent with our studies using *Pcdo-1::CDO-1::GFP* and *Pcdo-1::GFP* transgenes. Furthermore, high levels of cysteine also promoted accumulation of CDO-1::GFP (**Fig. S3**). In all scenarios, the hypodermis was the most prominent site of CDO-1::GFP accumulation. Specifically, we found that CDO-1::GFP was expressed in the cytoplasm of Hyp7, the major *C. elegans* hypodermal cell (**Fig. 4B**).

Based on the expression pattern of CDO-1::GFP encoded by *cdo-1(rae273)*, we hypothesized that CDO-1 acts in the hypodermis to promote sulfur amino acid metabolism. To test this hypothesis, we engineered a *cdo-1* rescue construct in which *cdo-1* is expressed exclusively in the hypodermis under the control of a hypoderm-specific collagen (*col-10*) promoter (*Pcol-10::CDO-1::GFP*) (53). Collagens are expressed exclusively in the hypodermis of nematodes, with a periodic induction in phase with the almost diurnal molting cycle (54). Multiple independent transgenic *C. elegans* strains were generated by integrating the *Pcol-10::CDO-1::GFP* construct into the *C. elegans* genome and tested for rescue of the *cdo-1(-)* mutant suppression of Moco-deficient larval arrest (41). Thus, tissue-specific complementation of the *cdo-1* mutation would regenerate a lethal arrest phenotype caused by Moco deficiency. We found that multiple independent transgenic strains of *cdo-1(-) moc-1(-)* double mutant animals carrying the *Pcol-10::CDO-1::GFP* transgene displayed a larval arrest phenotype when fed a Moco-deficient diet (**Fig. 4D**). These data demonstrated that hypodermal-specific expression of *cdo-1* is sufficient to rescue the *cdo-1(-)* mutant suppression of Moco-deficient lethality. This rescue was dependent upon the enzymatic activity of CDO-1 as an active site variant of this transgene (*Pcol-10::CDO-1[C85Y]::GFP)* did not rescue the suppressed larval arrest of *cdo-1(-) moc-1(-)* double mutant animals fed Moco-deficient diets (**Fig. 4D**). Taken together, our analyses of CDO-1::GFP expression demonstrate that CDO-1 is expressed, and that expression is regulated, in multiple tissues, principal among them being the hypodermis and Hyp7 cell. Our tissue-specific rescue data demonstrate that hypodermal expression of *cdo-1* is sufficient to promote cysteine catabolism and suggest that the hypodermis is a critical tissue for sulfur amino acid metabolism. However, we cannot exclude the possibility that CDO-1 also acts in other cells and tissues as well.

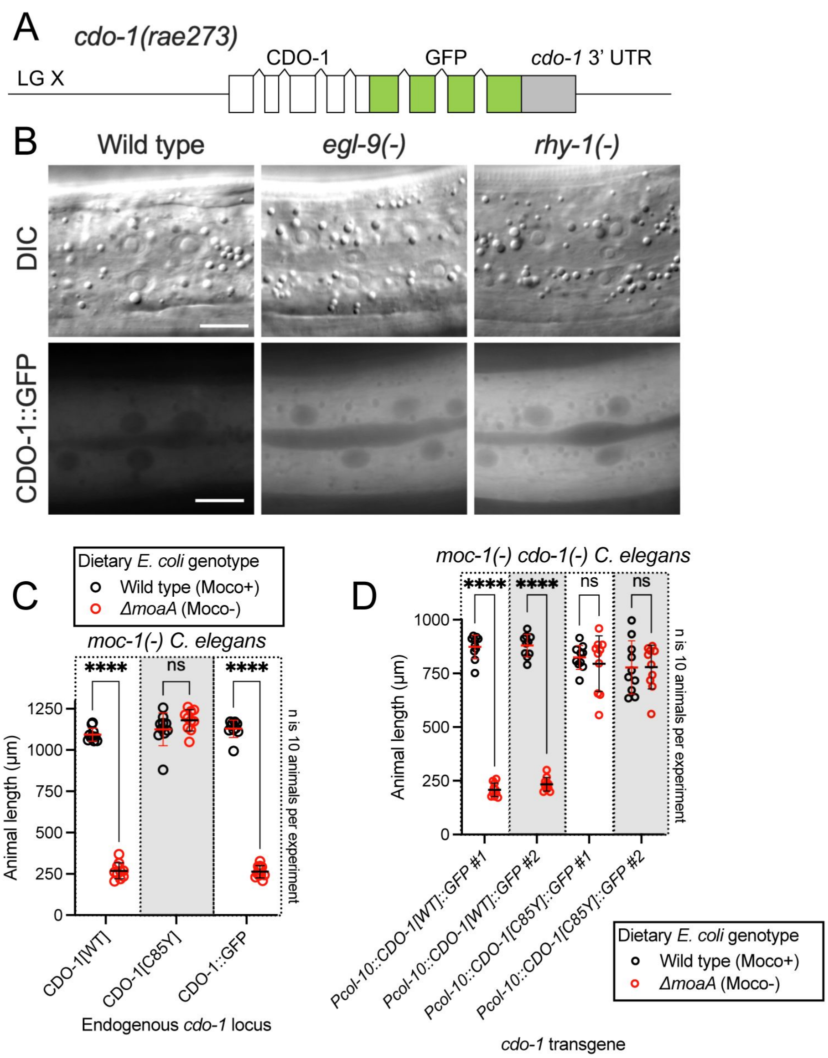

Figure 4

Hypodermal CDO-1 accumulates in the cytoplasm when *egl-9* or *rhy-1* are inactive and is sufficient to promote sulfur amino acid metabolism.

A) Diagram of *cdo-1(rae273),* a CRISPR/Cas9-generated allele with GFP inserted into the *cdo-1* gene, creating a functional C-terminal CDO-1::GFP fusion protein expressed from the native *cdo-1* locus. B) Differential interference contrast (DIC) and fluorescence imaging are shown for wild-type, *egl-9(sa307),* and *rhy-1(ok1402) C. elegans* expressing CDO-1::GFP encoded by *cdo-1(rae273).* Scale bar is 10µm. For GFP imaging, exposure time was 200ms. An anterior segment of the Hyp7 hypodermal cell is displayed. CDO-1::GFP accumulates in the cytoplasm and is excluded from the nuclei. C) *moc-1(ok366), moc-1(ok366) cdo-1(mg622),* and *moc-1(ok366) cdo-1(rae273)* animals were cultured from synchronized L1 larvae for 72 hours on wild-type (black, Moco+) or Δ*moaA* mutant (red, Moco-) *E. coli.* D) *moc-1(ok366) cdo-1(mg622)* double mutant animals expressing *Pcol-10::CDO-1::GFP* or *Pcol-10::CDO-1[C85Y]::GFP* transgenes were cultured for 48 hours on wild-type (black, Moco+) or Δ*moaA* mutant (red, Moco-) *E. coli.* Two independently derived strains were tested for each transgene. For panels C and D, animal lengths were determined for each condition. Individual datapoints are shown (circles) as are the mean and standard deviation. Sample size (*n*) is 10 individuals for each experiment. ****, p<0.0001, multiple unpaired t test with Welch's correction. ns indicates no significant difference was identified.

HIF-1 promotes CDO-1 activity downstream of the H_2S-sensing pathway

We sought to determine the physiological impact of *cdo-1* transcriptional activation by HIF-1. We reasoned mutations that activate HIF-1 and increase *cdo-1* transcription may cause increased CDO-1 activity. CDO-1 sits at a critical metabolic node in the degradation of the sulfur amino acids cysteine and methionine (**Fig 1A**). A key byproduct of sulfur amino acid metabolism and CDO-1 is sulfite, a reactive toxin that is detoxified by the Moco-requiring sulfite oxidase (SUOX-1). Null mutations in *suox-1* cause larval lethality. However, animals carrying the *suox-1(gk738847)* hypomorphic allele are healthy under standard culture conditions (33, 34). *suox-1(gk738847)* mutant animals display only 4% SUOX-1 activity compared to wild type and are exquisitely sensitive to sulfite stress (55). Thus, the *suox-1(gk738847)* mutation creates a sensitized genetic background to probe for increases in endogenous sulfite production. To test if increased *cdo-1* transcription would impact the growth of *suox-1*-comprimised animals, we combined the *egl-9* null mutation, which promotes HIF-1 activity and *cdo-1* transcription, with the *suox-1(gk738847)* allele. While *egl-9(-)* and *suox-1(gk738847)* single mutant animals are healthy under standard culture conditions, the *egl-9(-); suox-1(gk738847)* double mutant animals are extremely sick and require significantly more days to exhaust their *E. coli* food source under standard culture conditions (**Table 1**). These data establish a synthetic genetic interaction between these loci. To determine the role of sulfur amino acid metabolism in the *egl-9(-); suox-1(gk738847)* synthetic sickness phenotype, we engineered *egl-9(-); cdo-1(-) suox-1(gk738847)* and *cth-2(-); egl-9(-); suox-1(gk738847)* triple mutant animals. The *egl-9; suox-1* synthetic sickness phenotype was suppressed by inactivating mutations in *cdo-1* or *cth-2* which block the endogenous production of sulfite (**Table 1**). These data demonstrate that the deleterious activity of the *egl-9(-)* mutation in a *suox-1(gk738847)* background requires functional sulfur amino acid metabolism.

To determine the role of the H_2S-sensing pathway in the synthetic sickness phenotype displayed by *egl-9(-); suox-1(gk738847)* double mutant animals, we introduced null alleles of *cysl-1* or *hif-1* into the *egl-9(-); suox-1(gk738847)* double mutant. The *egl-9; suox-1* synthetic sickness phenotype was dependent upon *hif-1* but not *cysl-1* (**Table 1**). These results are consistent with our proposed genetic pathway and support the model that transcriptional activation of *cdo-1* by HIF-1 causes increased CDO-1 activity and increased flux of sulfur amino acids through their catabolic pathway.

Loss of *rhy-1* also strongly activates *cdo-1* transcription. We hypothesized that *rhy-1(-); suox-1(gk738847)* double mutant animals would display a synthetic sickness phenotype like *egl-9(-); suox-1(gk738847)* double mutant animals. However, based upon their ability to exhaust their *E. coli* food source under standard culture conditions, *rhy-1(-); suox-1(gk738847)* double mutant animals were just as healthy as either *rhy-1(-)* or *suox-1(gk738847)* single mutant *C. elegans* (**Table 1**). These data suggest that increasing *cdo-1* transcription alone is not sufficient to promote sulfite production via CDO-1. In addition to the role played by *rhy-1* in the regulation of HIF-1 activity, *rhy-1* itself is a transcriptional target of HIF-1. Loss of *egl-9* activity induces *rhy-1* mRNA ~50-fold in a *hif-1*-dependent manner (49). Given this potent transcriptional activation, we wondered if *rhy-1* might play an additional role downstream of HIF-1 in the regulation of sulfur amino acid metabolism and sulfite production. To test this hypothesis, we engineered *rhy-1(-); egl-9(-); suox-1(gk738847)* triple mutant animals. To our surprise, the *rhy-1(-); egl-9(-); suox-1(gk738847)* triple mutant animals were healthy, demonstrating that *rhy-1* was necessary for the deleterious activity of the *egl-9(-)* mutation in a *suox-1(gk738847)* background (**Table 1**). These genetic data suggest a dual role for *rhy-1* in the control of sulfur amino acid metabolism; first as a component of a regulatory cascade that controls the activity of HIF-1 and second as a functional downstream effector of HIF-1 that is required for sulfur amino acid metabolism.

This is not the first description of a *rhy-1* role downstream of *hif-1*. Overexpression of a *rhy-1*-encoding transgene suppresses the lethality of a *hif-1(-)* mutant during H_SS stress (56). These data establish RHY-1 as both a regulator and effector of HIF-1. How RHY-1, a predicted membrane-

C. elegans strain	Genetic locus 1	Genetic locus 2	Genetic locus 3	Days for population to starve petri dish ± SD (*n*)	Significant difference compared to GR2269 (Adjusted P Value)
N2	Wild type	Wild type	Wild type	5±1 (8)	No test
GR2269	Wild type	*suox-1(gk738847)*	Wild type	8±1 (4)	No test
JT307	*egl-9(sa307)*	Wild type	Wild type	7±1 (5)	No test
USD421	*egl-9(sa307)*	*suox-1(gk738847)*	Wild type	19±5 (11)	0.0003 (***)
USD926	*egl-9(rae276)*	Wild type	Wild type	6±1 (3)	No test
USD937	*egl-9(rae276)*	*suox-1(gk738847)*	Wild type	9±1 (3)	0.93 (ns)
USD512	*rhy-1(ok1402)*	Wild type	Wild type	6±0 (4)	No test
USD414	*rhy-1(ok1402)*	*suox-1(gk738847)*	Wild type	7±0 (4)	0.99 (ns)
CB5602	*vhl-1(ok161)*	Wild type	Wild type	6±0 (4)	No test
USD422	*vhl-1(ok161)*	*suox-1(gk738847)*	Wild type	8±1 (4)	0.99 (ns)
***C. elegans* strain**	**Genetic locus 1**	**Genetic locus 2**	**Genetic locus 3**	**Days for population to starve petri dish ± SD (n)**	**Significant difference compared to USD421 (Adjusted P value)**
USD421	*egl-9(sa307)*	*suox-1(gk738847)*	Wild type	19±5 (11)	No test
USD430	*egl-9(sa307)*	*suox-1(gk738847)*	*cdo-1(mg622)*	7±0 (6)	<0.0001 (****)
USD433	*egl-9(sa307)*	*suox-1(gk738847)*	*cth-2(mg599)*	9±1 (4)	<0.0001 (****)
USD432	*egl-9(sa307)*	*suox-1(gk738847)*	*rhy-1(ok1402)*	7±1 (4)	<0.0001 (****)
USD434	*egl-9(sa307)*	*suox-1(gk738847)*	*cysl-1(ok762)*	16±1 (9)	0.1 (ns)
USD431	*egl-9(sa307)*	*suox-1(gk738847)*	*hif-1(ia4)*	9±1 (6)	<0.0001 (****)

Table 1

Growth of *C. elegans* strains on standard laboratory conditions.

C. elegans strains and their corresponding mutations are displayed. For each strain, 5 L4-stage animals were seeded onto standard NGM petri dishes seeded with a monoculture of *E. coli* OP50. Petri dishes were monitored until the animals (and their progeny) depleted the lawn of *E. coli*. This was recorded as “days for population to starve petri dish”. The average of these experiments is displayed for each *C. elegans* strain as is the standard deviation (SD) and the number of biological replicates (*n*). Significant differences in population growth were determined by appropriate comparisons to either *suox-1(gk738847)* (GR2269) or *egl-9(sa307); suox-1(gk738847)* (USD421) using an ordinary one-way ANOVA with Dunnett’s post hoc analysis. No test indicates a statistical comparison was not made. Note, the data for USD421 are displayed twice in the table to allow for ease of comparison.

bound O-acyltransferase, molecularly executes these dual roles remains to be explored.

EGL-9 prolyl hydroxylase activity and VHL-1 are largely dispensable in the regulation of CDO-1

EGL-9 inhibits HIF-1 through its prolyl hydroxylase domain that hydroxylates HIF-1 proline 621 (**Fig. 5A**) (39). To evaluate the impact of the EGL-9 prolyl hydroxylase domain on the regulation of *cdo-1*, we generated a prolyl hydroxylase domain-inactive *egl-9* mutation using CRISPR/Cas9. We engineered an *egl-9* mutation that substitutes an alanine in place of histidine 487 (H487A) (**Fig. 1C**). Histidine 487 of EGL-9 is highly conserved and catalytically essential in the prolyl hydroxylase domain active site (**Fig. 5B**) (57, 58). To evaluate the impact of an inactive EGL-9 prolyl hydroxylase domain on the transcription of *cdo-1*, we engineered a *C. elegans* strain carrying the *egl-9(H487A)* mutation with the *Pcdo-1::GFP* transcriptional reporter. *egl-9(H487A)* caused a modest increase in *Pcdo-1::GFP* accumulation in the *C. elegans* intestine, suggesting that the EGL-9 prolyl hydroxylase domain is necessary to repress *cdo-1* transcription in the intestine (**Fig. 5C,D**). However, the activation of *Pcdo-1::GFP* by the *egl-9(H487A)* mutation was markedly less when compared to *Pcdo-1::GFP* activation caused by an *egl-9(-)* null mutation (**Fig. 5C,D**). These data suggest that EGL-9 has a prolyl hydroxylase domain-independent activity that is responsible for repressing *cdo-1* transcription.

EGL-9 hydroxylates specific proline residues on HIF-1. These hydroxylated proline residues are recognized by the von Hippel-Lindau E3 ubiquitin ligase (VHL-1). VHL-1-mediated ubiquitination promotes degradation of HIF-1 by the proteasome (**Fig. 5A**) (40). Thus, the EGL-9 prolyl hydroxylase domain and VHL-1 act in a pathway to regulate HIF-1. To determine the role of VHL-1 in regulating *cdo-1*, we engineered a *C. elegans* strain carrying the *vhl-1(-)* null mutation with our P*cdo-1::GFP* reporter. *vhl-1* inactivation also caused a modest increase in *Pcdo-1::GFP* expression in the *C. elegans* intestine, suggesting that *vhl-1* is necessary to repress *cdo-1* transcription (**Fig. 5C,D**). However, activation of *Pcdo-1::GFP* by the *vhl-1(-)* mutation was less than activation caused by an *egl-9(-)* null mutation (**Fig. 5C,D**). These data suggest that EGL-9 has a VHL-1-independent activity that is responsible for repressing *cdo-1* transcription.

To evaluate the impact of inactivating the EGL-9 prolyl hydroxylase domain or VHL-1 on cysteine metabolism, we again employed the *suox-1(gk738847)* hypomorphic mutation that sensitizes animals to increases in sulfite. We engineered *egl-9(H487A); suox-1(gk738847)* and *vhl-1(-) suox-1(gk738847)* double mutant animals and evaluated the health of those strains. In contrast to *egl-9(-); suox-1(gk738847)* double mutant animals which are extremely sick, *egl-9(H487A); suox-1(gk738847)* and *vhl-1(-) suox-1(gk738847)* double mutant animals are healthy (**Table 1**). These genetic data suggest that neither the EGL-9 prolyl hydroxylase domain nor VHL-1 are necessary to repress cysteine catabolism/sulfite production. However, it is plausible that the *egl-9(H487)* or *vhl-1(-)* mutations modestly activate cysteine metabolism, likely proportional to their activation of the *Pcdo-1::GFP* transgene (**Fig. 5C,D**), and that this activation is not sufficient to produce enough sulfites to negatively impact the growth of *suox-1(gk738847)* mutant animals.

Discussion

CDO-1 is a physiologically relevant effector of HIF-1

The hypoxia-inducible factor HIF-1 is a master regulator of the cellular response to hypoxia. It activates the transcription of many genes and pathways that are critical to maintain metabolic homeostasis in the face of low O_2. For example, mammalian HIF1α induces the hematopoietic growth hormone erythropoietin, glucose transport and glycolysis, as well as lactate dehydrogenase (59–62). The nematode *C. elegans* encounters a range of O_2 tensions in its natural habitat of rotting material: as microbial abundance increases, O_2 levels decrease from atmospheric levels

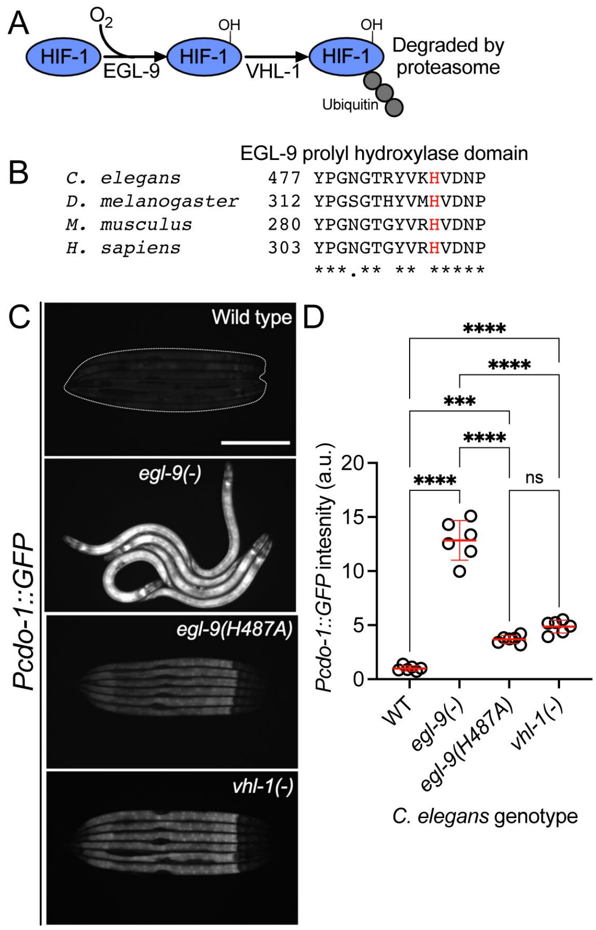

Figure 5

egl-9 inhibits *cdo-1* transcription in a largely prolyl-hydroxylase and VHL-1-independent manner.

A) The pathway of HIF-1 processing during normoxia is displayed. EGL-9 uses O_2 as a substrate to hydroxylate (-OH) HIF-1 on specific proline residues. Prolyl hydroxylated HIF-1 is bound by VHL-1 which facilitates HIF-1 polyubiquitination and targets HIF-1 for degradation by the proteasome. B) Amino acid alignment of the EGL-9 prolyl hydroxylase domain from *C. elegans, D. melanogaster, M. musculus,* and *H. sapiens.* "*" indicate perfect amino acid conservation while "." indicates weak similarity amongst species compared. Highlighted (red) is the catalytically essential histidine 487 residue in *C. elegans*. Alignment was performed using Clustal Omega (EMBL-EBI). C) Expression of *Pcdo-1::GFP* promoter fusion transgene is displayed for wild-type, *egl-9(sa307, -), egl-9(rae276,* H487A), and *vhl-1(ok161) C. elegans* animals at the L4 stage of development. Scale bar is 250µm. White dotted line outlines animals with basal GFP expression. For GFP imaging, exposure time was 500ms. D) Quantification of the data displayed in **Fig. 5C**. Individual datapoints are shown (circles) as are the mean and standard deviation (red lines). *n* is 6 individuals per genotype. Data are normalized so that wild-type expression of *Pcdo-1::GFP* is 1 arbitrary unit (a.u.). ***, p<0.001, ****, p<0.0001, ordinary one-way ANOVA with Tukey's multiple comparisons test. ns indicates no significant difference was identified.

(~21% O_2). In fact, *C. elegans* prefers 5-12% O_2, perhaps because hypoxia predicts abundant bacterial food sources (63). Members of the HIF-1 pathway and its targets have emerged from genetic studies of *C. elegans* (39). For instance, *C. elegans* HIF-1 promotes H_2S homeostasis by inducing transcription of the mitochondrial sulfide quinone oxidoreductase (*sqrd-1),* detoxifying H_2S (37).

We sought to define genes that regulate the levels and activity of cysteine dioxygenase (CDO-1), a critical regulator of cysteine homeostasis (14–16). Taking an unbiased genetic approach in *C. elegans,* we found that *cdo-1* was highly regulated by HIF-1 downstream of a signaling pathway that includes *rhy-1*, *cysl-1*, and *egl-9*. We demonstrated that HIF-1 promotes *cdo-1* transcription, accumulation of CDO-1 protein, and increased CDO-1 activity. Loss of *rhy-1* or *egl-9* activate *hif-1* to in turn strongly induce the *cdo-1* promoter. The pathway for activation of *cdo-1* also requires *cysl-1,* which functions downstream of *rhy-1* and upstream of *egl-9.* Based on ChIP-Seq studies, HIF-1 directly binds to the *cdo-1* promoter (50). Interestingly, mammalian CDO1 is regulated at both the transcriptional and post-transcriptional level in response to high dietary cysteine (17–21). Our studies in the nematode *C. elegans* suggest that CDO1 transcription in mammals might be governed by HIF1α in response to changes in cellular cysteine. Importantly, all members of the RHY-1/CYSL-1/EGL-9 pathway in *C. elegans* have homologs encoded by mammalian genomes (38). Given the conservation of these proteins, future studies may show that similar cysteine and H_2S-responsive signaling pathways operate in mammals.

We also show that the transcriptional activation of *cdo-1* by HIF-1 promotes CDO-1 enzymatic activity. Genetic activation of *cdo-1* by loss of *egl-9* causes severe sickness in a mutant with reduced sulfite oxidase activity, an activity required to cope with the toxic sulfite produced via CDO-1. This synthetic sickness is dependent upon an intact HIF-1 signaling pathway and a functioning sulfur amino acid metabolism pathway, as the dramatic sickness of an *egl-9; suox-1* double mutant is suppressed by loss of *hif-1*, *cth-2,* or *cdo-1*. These genetic results validate the intersection of HIF-1 and CDO-1 in converging biological regulatory pathways and encourage further exploration of this regulatory node. The potential physiological relevance of the connection between HIF-1 signaling and cysteine metabolism is discussed below.

It was initially surprising that the canonical hypoxia-sensing transcription factor, that is activated by relatively low O_2 tensions, induces transcription of *cdo-1,* which encodes an oxidase that requires dissolved O_2 to oxidize cysteine. Both EGL-9 and CDO-1 are dioxygenases, requiring O_2 as a substrate to catalyze their chemical reactions. EGL-9 functions as an O_2 sensor, using its O_2 substrate to hydroxylate HIF-1 at relatively high O_2 partial pressures, targeting it for degradation. EGL-9 is uniquely poised to sense deviations from the normally high physiological O_2 concentrations given its high K_m for O_2 (64–68). While it is not known if mammalian CDO1 or *C. elegans* CDO-1 dioxygenases are active at lower O_2 tensions than EGL-9, it is possible, even probable, that the set point for activation of HIF-1 by the failure of EGL-9 prolyl hydroxylation, is at a higher O_2 tension than the K_m of CDO-1 for oxidation of cysteine. In this way, CDO-1 could still coordinate cysteine catabolism while the cell is experiencing hypoxia and EGL-9 is unable to hydroxylate HIF-1.

Evidence for distinct pathways of HIF-1 activation by hypoxia and cysteine/H_2S

The hypoxia-signaling pathway is defined by EGL-9-dependent prolyl hydroxylation of HIF-1. Hydroxylated HIF-1 is then targeted for degradation via VHL-1-mediated ubiquitination (39, 40, 52). However, multiple lines of evidence, reinforced by our work, demonstrate VHL-1- and prolyl hydroxylase-independent activity of EGL-9. *egl-9(-)* null mutant *C. elegans* accumulate HIF-1 protein and display increased transcription of many genes, including *nhr-57,* an established target of HIF-1 (36). In rescue experiments of an *egl-9(-)* null mutant, Shao *et al.* (2009) demonstrate that a wild-type *egl-9* transgene restores normal HIF-1 protein levels and HIF-1 transcription. However, rescue experiments with a prolyl hydroxylase domain-inactive *egl-9(H487A)* transgene do not correct the accumulation of HIF-1 protein and only partially reduce the HIF-1 transcriptional output (58). Through our studies of *cdo-1* transcription, we demonstrate that an *egl-9(H487A)* mutant incompletely activates HIF-1 transcription when compared to an *egl-9(-)* null mutation (**Fig. 5C,D**). Taken together, we conclude EGL-9 has activity independent of its prolyl hydroxylase domain, mirroring and supporting previous work (58).

In studies of the HIF-1-dependent P*nhr-57::GFP* transcriptional reporter, Shen *et al.* (2006) observed that *egl-9(-)* null mutants promote HIF-1 transcription more than a *vhl-1(-)* mutant (36). We observe this same distinction between *egl-9* and *vhl-1* mutations with our *Pcdo-1::GFP* reporter (**Fig. 5C,D**). This difference in transcriptional activity is not explained by HIF-1 protein levels as HIF-1 protein accumulated equally in *egl-9(-)* and *vhl-1(-)* null mutants (36). This observation suggests that EGL-9 represses both HIF-1 levels and activity. This study also notes that *vhl-1* represses HIF-1 transcription in the intestine while *egl-9* acts in a wider array of tissues including the intestine and the hypodermis (36). Budde and Roth (2010) also demonstrate that loss of *vhl-1* promotes HIF-1 transcription in the *C. elegans* intestine while total loss of *egl-9* promotes HIF-1 transcription in multiple tissues including the intestine and the hypodermis (69). Our data expand upon these observations by demonstrating that loss of the EGL-9 prolyl hydroxylase domain promotes *cdo-1* transcription in the intestine alone, mirroring the loss of *vhl-1* (**Fig. 5C**). Budde and Roth (2010) additionally demonstrate that physiologically relevant stimuli also elicit a tissue-specific transcriptional response: hypoxia promotes HIF-1 transcription in the intestine while H_2S promotes HIF-1 transcription in the hypodermis. Importantly, H_2S promotes hypodermal HIF-1 transcription in a *vhl-1(-)* mutant, demonstrating a VHL-1-independent pathway for H_2S activation of HIF-1 (69). We demonstrate that high levels of cysteine promote *cdo-1* transcription in the hypodermis. Taken together with our data, these studies suggest two distinct pathways for activating HIF-1 transcription: i) a hypoxia-sensing pathway that is dependent upon *vhl-1* and the EGL-9 prolyl hydroxylase domain and promotes HIF-1 activity in the intestine and ii) an H_2S-sensing pathway that is independent of *vhl-1* and the EGL-9 prolyl hydroxylase domain and promotes HIF-1 activity in the hypodermis.

The focus of CDO-1 expression and regulation of sulfite production in the hypoderm may be due to the demand in sulfur metabolism as well as oxygen-dependent hydroxylation of collagens during the *C. elegans* molting cycle. Many collagen genes are expressed before each larval molt and many collagen prolines are hydroxylated, like particular HIF-1 prolines, and their cysteines form disulfides during collagen assembly. The large demand for the amino acid cysteine in the many collagen genes expressed at high levels during a molt may challenge cysteine homeostasis in the hypodermis (54).

The genetic details of the H_2S-sensing pathway were solidified through studies of *rhy-1* in *C. elegans.* Ma *et al.* (2012) demonstrate that *hif-1* repression via *rhy-1* requires the activity of *cysl-1*, a gene encoding a cysteine synthase-like protein (38). They further demonstrate that high H_2S promotes a physical interaction between CYSL-1 and EGL-9, resulting in the inactivation of EGL-9 and increased HIF-1 activity. Together, these studies suggest distinct genetic regulators of EGL-9/HIF-1 signaling: *rhy-1* and *cysl-1* govern the H_2S-sensing pathway while *vhl-1* mediates the hypoxia-sensing pathway. These pathways are distinct in their requirement for the EGL-9 prolyl hydroxylase domain.

A negative feedback loop senses high cysteine/H_2S, promotes CDO-1 activity, and maintains cysteine homeostasis

Why would the RHY-1/CYSL-1/EGL-9/HIF-1 H_2S-sensing pathway control the levels and activity of cysteine dioxygenase? We speculate this intersection facilitates a homeostatic pathway allowing *C. elegans* to sense and respond to cysteine level. We propose that H_2S acts as a gaseous signaling molecule to promote cysteine catabolism. H_2S activates HIF-1 in the hypodermis by promoting the

CYSL-1-mediated inactivation of EGL-9 (38). We show that high levels of cysteine similarly induce *cdo-1* transcription in the hypodermis. Our genetic data demonstrate that *cdo-1* is induced by the same genetic pathway that senses H_2S in *C. elegans* and CDO-1 acts in the hypodermis, the major site of H_2S-induced transcription. Furthermore, H_2S induces endogenous *cdo-1* transcription >3-fold while *cdo-1* mRNA levels do not change when *C. elegans* are exposed to hypoxia (70, 71). Thus, it is likely that H_2S promotes *cdo-1* transcription through RHY-1, CYSL-1, EGL-9, and HIF-1. H_2S is a reasonable small molecule signal to alert cells to high levels of cysteine. Excess cysteine results in the production of H_2S mediated by multiple enzymes including cystathionase (CTH), cystathionine β-synthase (CBS), and 3-mercaptopyruvate sulfurtransferase (MST) (9, 72). For example, CTH activity within the carotid body produces H_2S that modifies the mammalian response to hypoxia (73). We speculate that excess cysteine in *C. elegans* promotes the enzymatic production of H_2S which activates HIF-1 via the RHY-1/CYSL-1/EGL-9 signaling pathway. In our homeostatic model, H_2S-activated HIF-1 would then induce *cdo-1* transcription, promoting CDO-1 activity and the catabolism of the high-cysteine trigger (**Fig. 6**). Supporting this model, *cysl-1(-)* and *hif-1(-)* mutant *C. elegans* cannot survive in a high cysteine environment, demonstrating their central role in promoting cysteine homeostasis.

Importantly, the human ortholog of EGL-9 (EglN1) has previously been implicated as a cysteine sensor in the context of triple negative breast cancer (TNBC) (74). This study determined that HIF1α accumulates in TNBC cells even during normoxia. Briggs *et al.* (2016) demonstrate that L-glutamate secretion via TNBC cells suppresses HIF1α prolyl hydroxylation, stabilizing HIF1α. L-glutamate secretion is mediated via the glutamate/cystine antiporter, xCT (75, 76). L-glutamate secretion inhibits xCT and a concomitant decrease in intracellular cystine/cysteine was observed. The authors propose that low intracellular cystine/cysteine produces oxidizing conditions that oxidize specific EglN1 cysteine residues, inhibiting the activity of EglN1. Thus, Briggs *et al.* (2016) propose EglN1 as a cysteine sensor whose activity is promoted by cysteine.

Our work in *C. elegans* also strongly suggests a role for EGL-9 in sensing and responding to cysteine. We show that high levels of cysteine promote transcription of the HIF-1 target gene *cdo-1,* and that this induction is not additive with a null mutation in *egl-9,* suggesting cysteine and *egl-9* act in a pathway. Therefore, we propose that in *C. elegans* high levels of cysteine inhibit the activity of EGL-9, the opposite effect observed in TNBC cells. Furthermore, our genetic studies demonstrate that regulation of *cdo-1* transcription occurs largely independent of the EGL-9 prolyl hydroxylase domain and VHL-1, while the mechanism proposed by Briggs *et al.* (2016) suggests that the stabilization of HIF1α by L-glutamate is correlated with increased prolyl hydroxylation. Despite these differences, it seems likely that the roles for *C. elegans* EGL-9 and human EglN1 in sensing cysteine are connected. However, additional studies are required to determine if there is an evolutionary relationship between these cysteine-sensing mechanisms.

Members of the H_2S-sensing pathway have also been implicated in the *C. elegans* response to infection (44, 77). Many secreted proteins that mediate intercellular signaling or innate immune responses to pathogens are cysteine-rich and form disulfide bonds in the endoplasmic reticulum before protein secretion (78, 79). Levels of free cysteine may fall during the massive inductions of cysteine-rich secreted proteins in development or immune defense, as well as during the synthesis of collagens in the periodic molting cycle (54, 80). The oxidation of so many cysteines to disulfides in the endoplasmic reticulum might locally lower O_2 levels preventing EGL-9 hydroxylation of HIF-1. Thus, the intersection between HIF-1 and cysteine homeostasis we have uncovered may contribute to a regulatory axis in cell-cell signaling during development and in immune function.

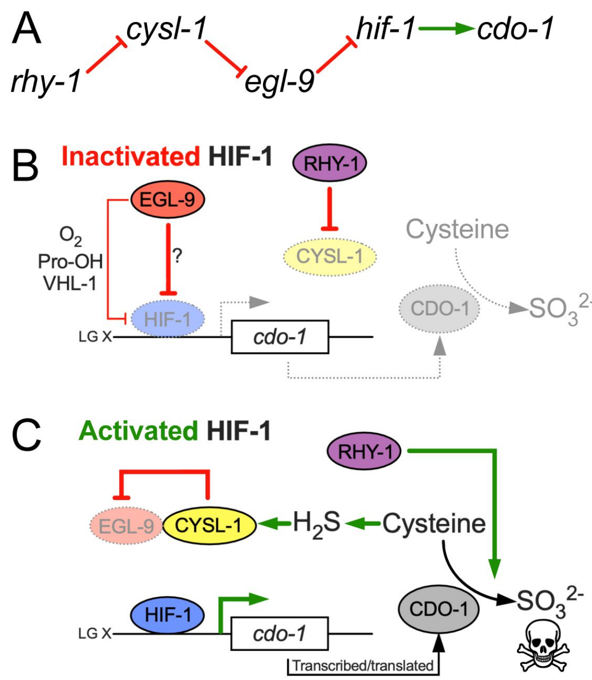

Figure 6

Model for the regulation of cysteine metabolism by HIF-1.

A) Proposed genetic pathway for the regulation of *cdo-1. rhy-1, cysl-1,* and *egl-9* act in a negative-regulatory cascade to control activity of the HIF-1 transcription factor, which activates transcription of *cdo-1.* B) Under basal conditions (inactivated HIF-1), EGL-9 negatively regulates HIF-1 through 2 distinct pathways; one pathway is dependent upon O_2, prolyl hydroxylation (Pro-OH), and VHL-1, while the second acts independently of these canonical factors. Under these conditions *cdo-1* transcription is kept at basal levels and cysteine catabolism is not induced. C) During conditions where HIF-1 is activated (high H_2S), CYSL-1 directly binds and inhibits EGL-9, preventing HIF-1 inactivation. Active HIF-1 binds the *cdo-1* promoter, driving transcription and promoting CDO-1 protein accumulation. High CDO-1 levels promote the catabolism of cysteine leading to production of sulfites (SO_3^{2-}) that are toxic during Moco or SUOX-1 deficiency. HIF-1-induced cysteine catabolism requires the activity of *rhy-1*.

Methods

General methods and strains

C. elegans were cultured using established protocols (43). Briefly, animals were cultured at 20°C on nematode growth media (NGM) seeded with wild-type *E. coli* (OP50). The wild-type strain of *C. elegans* was Bristol N2. Additional *E. coli* strains used in this work were BW25113 (Wild type, Moco+) and JW0764-2 (Δ*moaA753::kan,* Moco-) (81).

C. elegans mutant and transgenic strains used in this work are listed here. When previously published, sources of strains are referenced. Unless a reference is provided, all strains were generated in this study.

Non-transgenic *C. elegans*

- ZG31, *hif-1(ia4) V* (82)
- JT307, *egl-9(sa307) V* (44)
- GR2254, *moc-1(ok366) X* (33)
- GR2260, *cdo-1(mg622) X* (33)
- GR2261, *cdo-1(mg622) moc-1(ok366) X* (33)
- GR2269, *suox-1(gk738847) X* (33)
- CB5602, *vhl-1(ok161) X* (39)
- USD410, *cysl-1(ok762) X*, outcrossed 3x for this work
- USD414, *rhy-1(ok1402) II; suox-1(gk738847) X*
- USD421, *egl-9(sa307) V; suox-1(gk738847) X*
- USD422, *vhl-1(ok161) suox-1(gk738847) X*
- USD430, *egl-9(sa307) V; cdo-1(mg622) suox-1(gk738847) X*
- USD431, *hif-1(ia4) egl-9(sa307) V; suox-1(gk738847) X*
- USD432, *rhy-1(ok1402) II; egl-9(sa307) V; suox-1(gk738847) X*
- USD433, *cth-2(mg599) II; egl-9(sa307) V; suox-1(gk738847) X*
- USD434, *egl-9(sa307) X; cysl-1(ok762) suox-1(gk738847) X*
- USD512, *rhy-1(ok1402) II,* outcrossed 4x for this work
- USD706, *unc-119(ed3) III; cdo-1(mg622) moc-1(ok366) X*
- USD920, *cdo-1(rae273) moc-1(ok366) X*
- USD921, *egl-9(sa307) V; cdo-1(rae273) X*
- USD922, *rhy-1(ok1402) II; cdo-1(rae273) X*
- USD937, *egl-9(rae276) V; suox-1(gk738847) X*

MiniMos transgenic lines

- USD531, *unc-119(ed3) III; raeTi1 [Pcdo-1::CDO-1::GFP unc-119(+)]*
- USD719, *unc-119(ed3) III; cdo-1(mg622) moc-1(ok366); raeTi14 [Pcdo-1::CDO-1(C85Y)::GFP unc-119(+)]*
- USD720, *unc-119(ed3) III; raeTi15 [Pcdo-1::GFP unc-119(+)]*
- USD730, *rhy-1(ok1402) II; unc-119(ed3) III; raeTi15 [Pcdo-1::GFP unc-119(+)]*
- USD733, *unc-119(ed3) III; egl-9(sa307) V; raeTi15 [Pcdo-1::GFP unc-119(+)]*
- USD739, *unc-119(ed3) III; cdo-1(mg622) moc-1(ok366) X; raeTi1 [Pcdo-1::CDO-1::GFP unc-119(+)]*
- USD766, *unc-119(ed3) III; cdo-1(mg622) moc-1(ok366) X; raeTi32 [Pcol-10::CDO-1::GFP unc-119(+)]*
- USD767, *unc-119(ed3) III; cdo-1(mg622) moc-1(ok366) X; raeTi33 [Pcol-10::CDO-1::GFP unc-119(+)]*
- USD776, *rhy-1(ok1402) II; unc-119(ed3) III; hif-1(ia4) V; raeTi15 [Pcdo-1::GFP unc-119(+)]*
- USD777, *unc-119(ed3) III; egl-9(sa307) hif-1(ia4) V; raeTi15 [Pcdo-1::GFP unc-119(+)]*

- USD780, *rhy-1(ok1402) II; unc-119(ed3) III; cysl-1(ok762) X; raeTi15 [Pcdo-1::GFP unc-119(+)]*
- USD787, *unc-119(ed3) III; egl-9(sa307) V; cysl-1(ok762) X; raeTi15 [Pcdo-1::GFP unc-119(+)]*
- USD808, *unc-119(ed3) III; cdo-1(mg622) moc-1(ok366) X; raeTi40 [Pcol-10::CDO-1[C85Y]::GFP unc-119(+)]*
- USD810, *unc-119(ed3) III; cdo-1(mg622) moc-1(ok366) X; raeTi41 [Pcol-10::CDO-1[C85Y]::GFP unc-119(+)]*
- USD940, *unc-119(ed3) III; vhl-1(ok161) X; raeTi15 [Pcdo-1::GFP unc-119(+)]*
- USD1160, *unc-119(ed3) III; cysl-1(ok762) X; raeTi15 [Pcdo-1::GFP unc-119(+)]*
- USD1161, *unc-119(ed3) III; hif-1(ia4) V; raeTi15 [Pcdo-1::GFP unc-119(+)]*

EMS-derived strains

- USD659, *unc-119(ed3) III; egl-9(rae213) V; raeTi1 [Pcdo-1::CDO-1::GFP unc-119(+)]*
- USD674, *unc-119(ed3) III; egl-9(rae227) V; raeTi1 [Pcdo-1::CDO-1::GFP unc-119(+)]*
- USD655, *rhy-1(rae209) II; unc-119(ed3) III; raeTi1 [Pcdo-1::CDO-1::GFP unc-119(+)]*
- USD656, *rhy-1(rae210) II; unc-119(ed3) III; raeTi1 [Pcdo-1::CDO-1::GFP unc-119(+)]*
- USD657, *rhy-1(rae211) II; unc-119(ed3) III; raeTi1 [Pcdo-1::CDO-1::GFP unc-119(+)]*
- USD658, *rhy-1(rae212) II; unc-119(ed3) III; raeTi1 [Pcdo-1::CDO-1::GFP unc-119(+)]*

CRISPR/Cas9-derived strains

- USD914, *cdo-1(rae273) X,* CDO-1::GFP
- USD926, *egl-9(rae276) V,* EGL-9[H487A]
- USD928, *unc-119(ed3) III; egl-9(rae278) V; raeTi15 [Pcdo-1::GFP unc-119(+)]*

MiniMos transgenesis

Cloning of original plasmid constructs was performed using isothermal/Gibson assembly (83). All MiniMos constructs were assembled in pNL43, which is derived from pCFJ909, a gift from Erik Jorgensen (Addgene plasmid #44480) (84). Details about plasmid construction are described below. MiniMos transgenic animals were generated using established protocols that rescue the *unc-119(ed3)* Unc phenotype (41).

To generate a construct that expressed CDO-1 under the control of its native promoter, we cloned the wild-type *cdo-1* genomic locus from 1,335 base pairs upstream of the *cdo-1* ATG start codon to (and including) codon 190 encoding the final CDO-1 amino acid prior to the TAA stop codon. This wild-type genomic sequence was fused in frame with a C-terminal GFP and *tbb-2* 3'UTR (376 bp downstream of the *tbb-2* stop codon) in the pNL43 plasmid backbone. This plasmid is called pKW24 (P*cdo-1::CDO-1::GFP*).

To generate pKW44, a construct encoding the active site mutant transgene *Pcdo-1::CDO-1(C85Y)::GFP*, we performed Q5 site-directed mutagenesis on pKW24, following manufacturer's instructions (New England Biolabs). In pKW44, codon 85 was mutated from TGC (cysteine) to TAC (tyrosine).

To generate pKW45 (P*cdo-1::GFP*), the 1,335 base pair *cdo-1* promoter was amplified and fused directly to the GFP coding sequence. Both fragments were derived from pKW24, excluding the *cdo-1* coding sequence and introns.

pKW49 is a construct driving *cdo-1* expression from the hypodermal-specific *col-10* promoter (*Pcol-10::CDO-1::GFP*) (53). The *col-10* promoter (1,126 base pairs upstream of the *col-10* start codon) was amplified and fused upstream of the *cdo-1* ATG start codon in pKW24, replacing the native *cdo-1* promoter. pKW53 [P*col-10::CDO-1(C85Y)::GFP*] was engineered using the same Q5-site-directed mutagenesis strategy as was described for pKW44. However, this mutagenesis used pKW49 as the template plasmid.

Chemical mutagenesis and whole genome sequencing

To define *C. elegans* gene activities that were necessary for the control of *cdo-1* levels, we carried out a chemical mutagenesis screen for animals that accumulate CDO-1 protein. To visualize CDO-1 levels, we engineered USD531, a transgenic *C. elegans* strain carrying the *raeTi1* [pKW24, *Pcdo-1::CDO-1::GFP*] transgene. USD531 transgenic *C. elegans* were mutagenized with ethyl methanesulfonate (EMS) using established protocols (43). F2 generation animals were manually screened, and mutant isolates were collected that displayed high accumulation of *Pcdo-1::CDO-1::GFP*. We demanded that mutant strains of interest were viable and fertile.

We followed established protocols to identify EMS-induced mutations in our strains of interest (85). Briefly, whole genomic DNA was prepared from *C. elegans* using the Gentra Puregene Tissue Kit (Qiagen) and genomic DNA libraries were prepared using the NEBNext genomic DNA library construction kit (New England Biolabs). DNA libraries were sequenced on an Illumina Hi-Seq and deep sequencing reads were analyzed using standard methods on Galaxy, a web-based platform for computational analyses (86). Briefly, sequencing reads were trimmed and aligned to the WBcel235 *C. elegans* reference genome (87, 88). Variations from the reference genome and the putative impact of those variations were annotated and extracted for analysis (89–91). Here we report the analysis of 6 new mutant strains (USD655, USD656, USD657, USD658, USD659, and USD674). Among these mutant strains, we found 2 unique mutations in *egl-9* (USD659 and USD674) and 4 unique mutations in *rhy-1* (USD655, USD656, USD657, USD658). The allele names and molecular identity of these new *egl-9* and *rhy-1* mutations are specified in **Fig. 1C**. These genes were prioritized based on the isolation of multiple independent alleles and their established functions in a common pathway, the hypoxia and H_2S-sensing pathway (36–38).

Genome engineering by CRISPR/Cas9

We followed standard protocols to perform CRISPR/Cas9 genome engineering of *cdo-1* and *egl-9* genomic loci in *C. elegans* (92–95). Essential details of the CRISPR/Cas9-generated reagents in this work are described below.

We used homology-directed repair to generate *cdo-1(rae273)* [CDO-1::GFP]. The guide RNA was 5'-gactacagaggatctaagaa-3' (crRNA, IDT). The GFP donor double-stranded DNA (dsDNA) was amplified from pCFJ2249 using primers that contained roughly 40bp of homology to *cdo-1* flanking the desired insertion site (96). The primers used to generate the donor dsDNA were: 5'-gtacggcaagaaagttgactacagaggatctaagaataatagtactagcggtggcagtgg-3' and 5'-agaatcaacacgttattacattgagggatatgttgtttacttgtagagctcgtccattcc-3'. Glycine 189 of CDO-1 was removed by design to eliminate the PAM site and prevent cleavage of the donor dsDNA.

The *egl-9(rae276)* and *egl-9(rae278)* [EGL-9(H487A)] alleles were also generated by homology-directed repair using the same combination of guide RNA and single-stranded oligodeoxynucleotide (ssODN) donor. The guide RNA was 5'-tgtgaagcatgtagataatc-3' (crRNA, IDT) The ssODN donor was 5'-gcttgccatctatcctggaaatggaactcgttatgtgaaggctgtagacaatccagtaaaagatggaagatgtataaccactatttattactg-3' (Ultramer, IDT). Successful editing resulted in altering the coding sequence of EGL-9 to encode for an alanine rather than the catalytically essential histidine at position 487. We also used synonymous mutations to introduce an AccI restriction site that is helpful for genotyping the *rae276* and *rae278* mutant alleles.

C. elegans growth assays

To assay developmental rates, *C. elegans* were synchronized at the first stage of larval development. To synchronize animals, embryos were harvested from gravid adult animals via treatment with a bleach and sodium hydroxide solution. Embryos were then incubated overnight in M9 solution causing them to hatch and arrest development at the L1 stage (97). Synchronized L1 animals were cultured on NGM seeded with BW25113 (Wild type, Moco+) or JW0764-2 (Δ*moaA753::kan,* Moco-) *E. coli.* Animals were cultured for 48 or 72 hours (specified in the appropriate figure legends), and live animals were imaged as described below. Animal length was measured from tip of head to the end of the tail.

To determine qualitative 'health' of various *C. elegans* strains, we assayed the ability of these strains to consume all *E. coli* food provided on an NGM petri dish. For this experiment, dietary *E. coli* was produced via overnight culture in liquid LB in a 37°C shaking incubator. 200µl of this *E. coli* was seeded onto NGM petri dishes and allowed to dry, producing nearly identical lawns and growth environments. Then, 5 L4 animals of a strain of interest were introduced onto these NGM petri dishes seeded with OP50. For each experiment, petri dishes were monitored daily and scored when all *E. coli* was consumed by the population of animals. This assay is beneficial because it integrates many life-history measures (i.e. developmental rate, brood size, embryonic viability, etc.) into a single simple assay that can be scaled and applied to many *C. elegans* strains in parallel.

Cysteine exposure

To determine the impact of supplemental cysteine on expression of *cdo-1* and animal viability, we exposed various *C. elegans* strains to 0, 50, 100, 250, 500, or 1,000µM supplemental cysteine. *C. elegans* strains were synchronously grown on NGM media supplemented with *E. coli* OP50 at 20°C until reaching the L4 stage of development. Live L4 animals were then collected from the petri dishes and cultured in liquid M9 media containing 4X concentrated *E. coli* OP50 with or without supplemental cysteine. These liquid cultures were gently rocked at 20°C overnight. Post exposure, GFP imaging was performed as described in the Microscopy section of the materials and methods. Alternatively, animals exposed overnight to 0, 100, or 1,000µM supplemental cysteine were scored for viability after being seeded onto NGM petri dishes. Animals were determined to be alive if they responded to mechanical stimulus.

Microscopy

Low magnification bright field and fluorescence images (imaging GFP simultaneously in multiple animals) were collected using a Zeiss AxioZoom V16 microscope equipped with a Hamamatsu Orca flash 4.0 digital camera using Zen software (Zeiss). For experiments with supplemental cysteine, low magnification bright field and fluorescence images were collected using a Nikon SMZ25 microscope equipped with a Hamamatsu Orca flash 4.0 digital camera using NIS-Elements software (Nikon). High magnification differential interference contrast (DIC) and GFP fluorescence images (imaging CDO-1::GFP encoded by *cdo-1(rae273)*) were collected using Zeiss AxioImager Z1 microscope equipped with a Zeiss AxioCam HRc digital camera using Zen software (Zeiss). All images were processed and analyzed using ImageJ software (NIH). All imaging was performed on live animals paralyzed using 20mM sodium azide. For all fluorescence images shown within the same figure panel, images were collected using the same exposure time and processed identically. To quantify GFP expression, the average pixel intensity was determined within a set transverse section immediately anterior to the developing vulva. Background pixel intensity was determined in a set region of interest distinct from the *C. elegans* samples and was subtracted from the sample measurements.

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Supplemental Figures

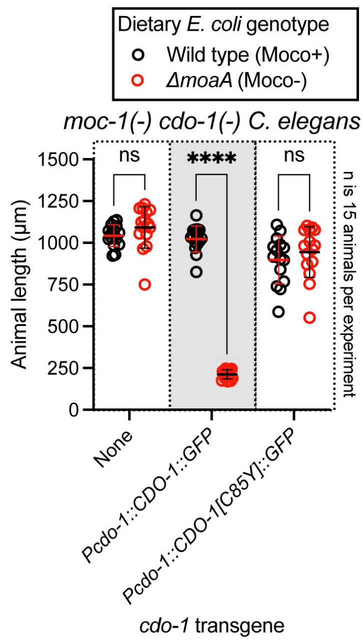

Figure S1

The *Pcdo-1::CDO-1::GFP* transgene encodes a functional cysteine dioxygenase enzyme.

moc-1(ok366) cdo-1(mg622) double mutant animals expressing *Pcdo-1::CDO-1::GFP* or *Pcdo-1::CDO-1[C85Y]::GFP* transgenes were cultured from synchronized L1 larvae for 72 hours on wild-type (black, Moco+) or Δ*moaA* mutant (red, Moco-) *E. coli.* Animal lengths were determined for each condition. Individual datapoints are shown (circles) as are the mean and standard deviation. Sample size (*n*) is displayed for each experiment. ****, p<0.0001, multiple unpaired t test with Welch's correction. ns indicates no significant difference was identified.

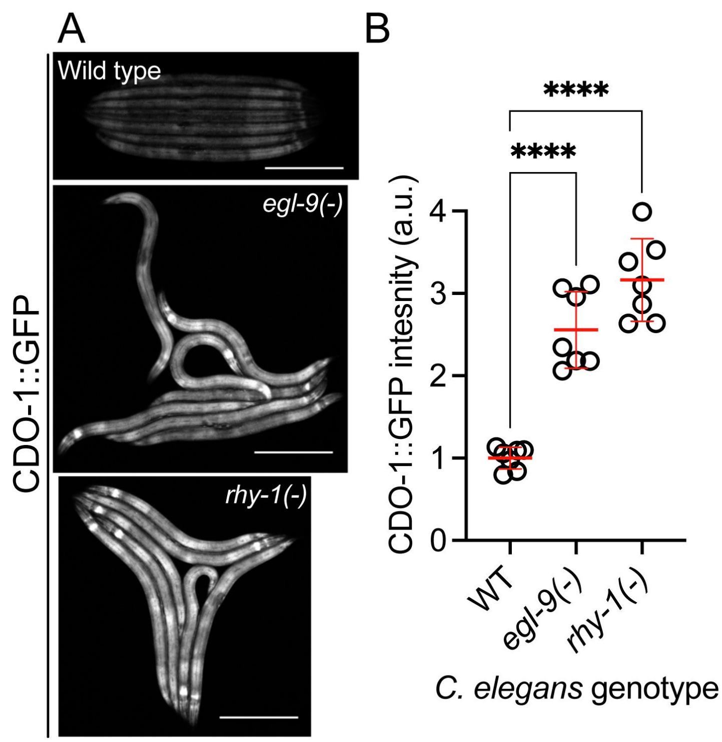

Figure S2

A functional CDO-1::GFP fusion protein is induced by loss of *egl-9* or *rhy-1*.

A) Expression of CDO-1::GFP from the *cdo-1(rae273)* allele is displayed for wild-type, *egl-9(sa307),* and *rhy-1(ok1402)* animals at the L4 stage of development. Scale bar is 250µm. For GFP imaging, exposure time was 100ms. B) Quantification of CDO-1::GFP expression displayed in **Fig. S2A**. Individual datapoints are shown (circles) as are the mean and standard deviation (red lines). *n* is 7 individuals per genotype. Data are normalized so that wild-type expression of CDO-1::GFP is 1 arbitrary unit (a.u.). ****, p<0.0001, ordinary one-way ANOVA with Dunnett's post hoc analysis.

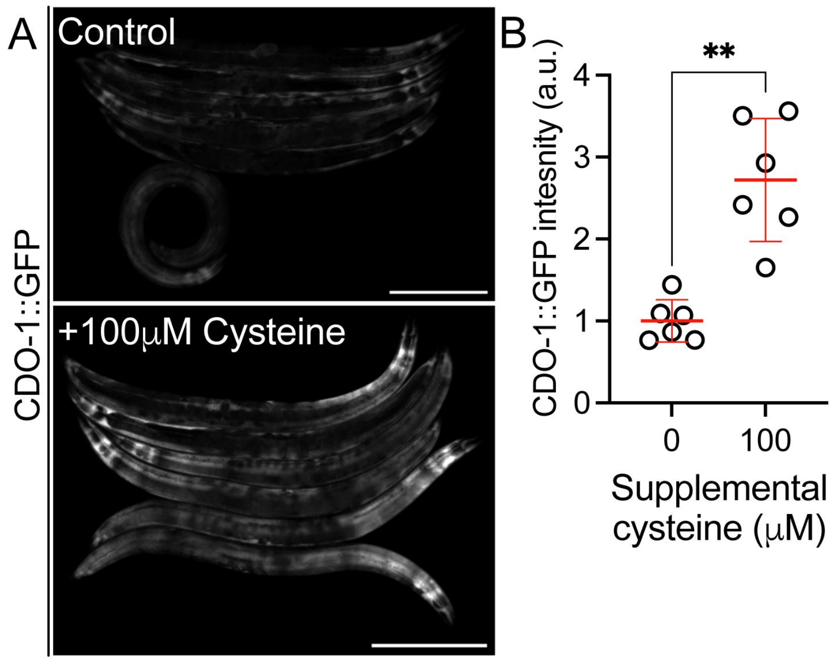

Figure S3

CDO-1::GFP encoded by *cdo-1(rae273)* is induced by supplemental cysteine.

A) Expression of CDO-1::GFP from the *cdo-1(rae273)* allele is displayed for wild-type young adult animals exposed to 0 or 100μM supplemental cysteine. Scale bar is 250μm. For GFP imaging, exposure time was 100ms. Supplemental cysteine did not impact the mortality of the animals being imaged. B) Quantification of CDO-1::GFP expression displayed in **Fig. S3A**. Individual datapoints are shown (circles) as are the mean and standard deviation (red lines). *n* is 6 individuals per genotype. Data are normalized so that wild-type expression of CDO-1::GFP exposed to 0μM supplemental cysteine is 1 arbitrary unit (a.u.). **, $p<0.001$, unpaired t test with Welch's correction.

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Reviewer #2 (Public Review):

The authors investigate the transcriptional regulation of cysteine dioxygenase (CDO-1) in C. elegans and its role in maintaining cysteine homeostasis. They show that high cysteine levels activate cdo-1 transcription through the hypoxia-inducible transcription factor HIF-1. Using transcriptional and translational reporters for CDO-1, the authors propose that a negative feedback pathway involving RHY-1, CYSL-1, EGL-9 and HIF-1 in regulating cysteine homeostasis.

Genetics is a notable strength of this study. The forward genetic screen, gene interaction and epistasis analyses are beautifully designed and rigorously conducted, yielding solid and unambiguous conclusions on the genetic pathway regulating CDO-1. The writing is clear and accessible, contributing to the overall high quality of the manuscript.
Addressing the specifics of cysteine supplementation and interpretation regarding the cysteine homeostasis pathway would further clarify the paper and strengthen the study's conclusions.

First, the authors show that the supplementation of exogenous cysteine activates cdo-1p::GFP. Rather than showing data for one dose, the author may consider presenting dose-dependency results and whether cysteine activation of cdo-1 also requires HIF-1 or CYSL-1, which would be important data given the focus and major novelty of the paper in cysteine homeostasis, not the cdo-1 regulatory gene pathway. While the genetic manipulation of cdo-1 regulators yields much more striking results, the effect size of exogenous cysteine is rather small. Does this reflect a lack of extensive condition optimization or robust buffering of exogenous/dietary cysteine? Would genetic manipulation to alter intracellular cysteine or its precursors yield similar or stronger effect sizes?

Second, there remain several major questions regarding the interpretation of the cysteine homeostasis pathway. How much specificity is involved for the RHY-1/CYSL-1/EGL-9/HIF-1 pathway to control cysteine homeostasis? Is the pathway able to sense cysteine directly or indirectly through its metabolites or redox status in general? Given the very low and high physiological concentrations of intracellular cysteine and glutathione (GSH, a major reserve for cysteine), respectively, there is a surprising lack of mention and testing of GSH metabolism. In addition, what are the major similarities and differences of cysteine homeostasis pathways between C. elegans and other systems (HIF dependency, transcription vs post-transcriptional control)? These questions could be better discussed and noted with novel findings of the current study that are likely C. elegans specific or broadly conserved.

All of my comments and questions above have been satisfactorily addressed in the revised manuscript.

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Reviewer #3 (Public Review):

There has been a long-standing link between the biology of sulfur-containing molecules (e.g., hydrogen sulfide gas, the amino acid cysteine, and its close relative cystine, et cetera) and the biology of hypoxia, yet we have a poor understanding of how and why these two biological processes and are co-regulated. Here, the authors use C. elegans to explore the relationship between sulfur metabolism and hypoxia, examining the regulation of cysteine dioxygenase (CDO1 in humans, CDO-1 in C. elegans), which is critical to cysteine catabolism, by the hypoxia inducible factor (HIF1 alpha in humans, HIF-1 in C. elegans), which is the key terminal effector of the hypoxia response pathway that maintains oxygen homeostasis. The authors are trying to demonstrate that (1) the hypoxia response pathway is a key regulator of cysteine homeostasis, specifically through the regulation of cysteine dioxygenase, and (2) that the pathway responds to changes in cysteine homeostasis in a mechanistically distinct way from how it responds to hypoxic stress.

Briefly summarized here, the authors initiated this study by generating transgenic animals expressing a CDO-1::GFP protein chimera from the cdo-1 promoter so that they could identify regulators of CDO-1 expression through a forward genetic screen. This screen identified mutants with elevated CDO-1::GFP expression in two genes, egl-9 and rhy-1, whose wild-type products are negative regulators of HIF-1, raising the possibility that cdo-1 is a HIF-1 transcriptional target. Indeed, the authors provide data showing that cdo-1 regulation by

EGL-9 and RHY-1 is dependent on HIF-1 and that regulation by RHY-1 is dependent on CYSL-1, as expected from other published findings of this pathway. The authors show that exogenous cysteine activates cdo-1 expression, reflective of what is known to occur in other systems. Moreover, they find that exogenous cysteine is toxic to worms lacking CYSL-1 or HIF-1 activity, but not CDO-1 activity, suggesting that HIF-1 mediates a survival response to toxic levels of cysteine and that this response requires more than just the regulation of CDO-1. The authors validate their expression studies using a GFP knockin at the cdo-1 locus, and they demonstrate that a key site of action for CDO-1 is the hypodermis. They present genetic epistasis analysis supporting a role for RHY-1, both as a regulator of HIF-1 and as a transcriptional target of HIF-1, in offsetting toxicity from aberrant sulfur metabolism. The authors use CRISPR/Cas9 editing to mutate a key amino acid in the prolyl hydroxylase domain of EGL-9, arguing that EGL-9 inhibits CDO-1 expression through a mechanism that is largely independent of the prolyl hydroxylase activity.

Overall, the data seem rigorous, and the conclusions drawn from the data seem appropriate. The experiments test the hypothesis using logical and clever molecular genetic tools and design. The sample size is a bit lower than is typical for C. elegans papers; however, the experiments are clearly not underpowered, so this is not an issue. The paper is likely to drive many in the field (including the authors themselves) into deeper experiments on (1) how the pathway senses hypoxia and sulfur/cysteine/H2S using these distinct mechanisms/modalities, (2) how oxygen and sulfur/cysteine/H2S homeostasis influence one another, and (3) how this single pathway evolved to sense and respond to both of these stress modalities.

My previous concerns have been addressed. The authors are commended on an excellent body of research.

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Reviewer #4 (Public Review):

Summary:
This is a revised manuscript that describes a role for cdo-1 in regulating cellular cysteine levels. The authors show that expression of cdo-1, predicted to encode a cysteine dioxygenase, is regulated by HIF-1, the conserved hypoxia-induced transcription factor. The expression of cdo-1 is controlled by the RHY-1/CYSL-1/EGL-9/HIF-1 pathway that has been demonstrated to be involved in the response to H2S.

Strengths:
The new finding of this study is that cdo-1, predicted to encode a cysteine dioxygenase, is expressed in the hypodermis and that hypodermal expression rescues at least one phenotype of the cdo-1(mg622) mutant (ability to survive toxic sulfite accumulation in Moco-deficient conditions). Using sulfite toxicity is an interesting reporter for cellular cysteine abundance.

Weaknesses:
The authors claim more than once that the H2S/Cys responsive pathway is RHY-1 - CYSL-1 - EGL-9 - HIF-1. Their data don't seem to support this claim, as they show that Pcdo-1::GFP is induced in rhy-1 mutants incubated with cysteine. It is therefore not appropriate to claim that "HIF-1-induced cysteine catabolism requires the activity of rhy-1" that they include in the description of the model in Fig 6. There is simply no evidence at all that RHY-1 has any role in modulating the activity of CDO-1 other than through transcriptional activation via HIF-1.

I don't find the arguments that this pathway is required for cysteine homeostasis per se (as claimed in the last sentence of the introduction). The authors expose worms to excess cysteine for 48 hours in liquid culture with bacteria. It is well known in these conditions that the bacteria will produce H2S from the cysteine in the culture. All of the cysteine exposure data shown can be explained by the effect of H2S exposure. This would explain why hif-1 and cysl-1 mutants die but cdo-1 mutants do not, for example. The authors don't provide any data to rule out the possibility that bacterial H2S production underlies these results. This explains why the pathway described in this work is the same as has been previously described. Similarly, there is no evidence at all to support their assertion that there are "other pathways" induced by HIF-1 to deal with sulfite produced by cysteine catabolism. However, if the main problem is H2S production (perhaps by bacteria) then cdo-1 would not be relevant and the mutants would be viable as observed.

In a couple of places, the authors seem to argue that H2S-induced expression is limited to the hypodermis and hypoxia-induced gene expression is mostly in the intestine. This is consistent with the expression of cdo-1 (this work) and nhr-57 (Budde and Roth) but it is not appropriate to generalize this. Previous work from the Ruvkun lab (Ma et al) show that the CYSL-1 regulates expression of HIF-1 targets in neurons. Moreover, HIF-1 protein accumulates in the nucleus of nearly all cells, and there is no reason to believe that there are changes in the expression of other genes in different tissues.

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Author Response

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Public Review):***
>
> *Issue 1: The relevance is somewhat unclear. High cysteine levels can be achieved in the laboratory, but, is this relevant in the life of C. elegans? Or is there physiological relevance in humans, e.g. a disease? The authors state "cells and animals fed excess cysteine and methionine", but is this more than a laboratory excess condition? SUOX nonfunctional conditions in humans don't appear to tie into this, since, in that context, the goal is to inactivate CDO or CTH to prevent sulfite production. The authors also mention cancer, but the link to cysteine levels is unclear. In that sense, then, the conditions studied here may not carry much physiological relevance.*

Response 1: We set out to answer a fundamental question: what pathways regulate the function of cysteine dioxygenase, a highly conserved enzyme in sulfur amino acid metabolism? In an unbiased genetic screen that sampled millions of EMS generated mutations across all ~20,000 C. elegans genes, we discovered loss of function/null mutations in egl-9 and rhy-1, two negative regulators of the hypoxia inducible transcription factor (hif-1). Genetic ablation of the egl-9 or rhy-1 loci are likely not relevant to the life of a C. elegans animal, i.e. this is not representative of a natural state. Yet, this extreme genetic intervention has taught us a new fundamental truth about the interaction between EGL-9/RHY-1, HIF-1, and the transcriptional activation of cdo1. Similarly, the high cysteine levels used in our assays may or may not be representative of a state in nature, we do not know (nor do we make any claims about the environmental relevance of our choice of cysteine concentrations). It seems very plausible that pathological states exist where cysteine concentrations may rise to comparable levels in our experimental system. More importantly, we have started with excess to physiology to elicit a clear response that we can study in the lab. Similar strategies established the cysteine-induction phenotype of CDO1 in mammalian systems. For instance, in Kwon and Stipanuk 2001, hepatocytes are cultured in media supplemented with 2mmol/L cysteine to promote a ~4-fold increase in CDO1 mRNA.

> *Issue 2: The pathway is described as important for cysteine detoxification, which is described to act via H2S (Figure 6). Much of that pathway has already been previously established by the Roth, Miller, and Horvitz labs as critical for the H2S response. While the present manuscript adds some additional insight such as the additional role of RHY-1*

> *downstream on HIF-1 in promoting toxicity, this study therefore mainly confirms the importance of a previously described signalling pathway, essentially adding a new downstream target rhy-1 -> cysl-1 -> egl-9 -> hif-1 -> sqrd-1/cdo-1. The impact of this finding is reduced by the fact that cdo-1 itself isn't actually required for survival in high cysteine, suggesting it is merely a maker of the activity of this previously described pathway.*

Response 2: We agree that the primary impact of our manuscript is the establishment of a novel intersection between the H2S-sensing pathway (largely worked out by Roth, Miller, and Horvitz) and our gene of interest, cysteine dioxygenase. We believe that the connection between these two pathways is exciting as it suggests a logical homeostatic circuit. High cysteine yields enzymatically produced H2S. This H2S may then act as a signal promoting HIF-1 activity (via RHY-1/CYSL-1/EGL-9). High HIF-1 activity increases cdo-1 transcription and activity promoting the degradation of the high-cysteine trigger. As pointed out by the reviewer, cdo-1(-) loss of function alone does not cause cysteine sensitivity at the concentrations tested. Given that cysl-1(-) and hif-1(-) mutants are exquisitely sensitive to high levels of cysteine, we propose that HIF-1 activates the transcription of additional genes that are required for high cysteine tolerance. However, our genetic data show that cdo-1 is more than simply a marker of HIF-1 transcription. Our genetic data in Table 1 demonstrate that HIF-1 activation (caused by egl-9(-)) is sufficient to cause severe sickness in a suox-1 hypomorphic mutant which cannot detoxify sulfites, a critical product of cysteine catabolism. This severe sickness can be reversed by inactivating hif-1, cth-2, or cdo-1. These data demonstrate a functional intersection between the established H2S-sensing pathway and cysteine catabolism governed by cdo-1.

> ***Reviewer #2 (Public Review):***
>
> *Issue 3: First, the authors show that the supplementation of exogenous cysteine activates cdo-1p::GFP. Rather than showing data for one dose, the author may consider presenting dose-dependency results and whether cysteine activation of cdo-1 also requires HIF-1 or CYSL-1, which would be important data given the focus and major novelty of the paper in cysteine homeostasis, not the cdo-1 regulatory gene pathway.*

Response 3: We agree with the reviewer and have performed the suggested dose-response curve for expression of Pcdo-1::GFP in wild-type C. elegans. We observe substantial activation of the Pcdo-1::GFP transcriptional reporter beginning at 100µM supplemental cysteine (Figure 3C). Higher doses of cysteine do not elicit a substantially stronger induction of the Pcdo-1::GFP reporter. Thus, we find that 100µM supplemental cysteine strikes the right balance between strongly inducing the Pcdo-1::GFP reporter while not inducing any toxicity or lethality in wild-type animals (Figure 3E).

We further agree that testing for induction of the Pcdo-1::GFP reporter in a hif-1(-) or cysl-1(-) mutant background is a critical experiment. However, we have not been able to identify a cysteine concentration that induces Pcdo-1::GFP and is not 100% lethal for hif-1(-) or cysl-1(-) mutant C. elegans. The remarkable sensitivity of hif-1(-) or cysl-1(-) mutant C. elegans to supplemental cysteine demonstrates the critical role of these genes in promoting cysteine homeostasis. But because of this lethality, we could not assay the Pcdo1::GFP reporter in the hif-1(-) or cysl-1(-) mutant animals. But the lethality to excess cysteine demonstrates that this cysteine response is salient. To get at how cysteine might be interacting with the HIF-1-signaling pathway, we performed new additivity experiments by supplementing 100µM cysteine to wild type, egl-9(-), and rhy-1(-) mutant C. elegans expressing the Pcdo-1::GFP reporter. Surprisingly, we found that cysteine had no significant impact on Pcdo-1::GFP expression in an egl-9(-) mutant background but significantly increased the Pcdo-1::GFP expression in a rhy-1(-) background (Figure 3A,B). These data suggest that cysteine acts in a pathway with egl-9 and in parallel to rhy-1. These data have been incorporated into Figure 3A,B and are included in the Results section of the manuscript.

> *Issue 4: While the genetic manipulation of cdo-1 regulators yields much more striking results, the effect size of exogenous cysteine is rather small. Does this reflect a lack of extensive condition optimization or robust buffering of exogenous/dietary cysteine? Would genetic manipulation to alter intracellular cysteine or its precursors yield similar or stronger effect sizes?*

Response 4: We agree that the induction of the Pcdo-1::GFP reporter by supplemental cysteine is not as dramatic as the induction caused by the egl-9 or rhy-1 null alleles. We believe our Response 3 and new Figure 3C demonstrate that this phenomenon is not due to lack of condition optimization, but likely reflects some biology. As pointed out by the reviewer, C. elegans likely buffers exogenous cysteine and this (perhaps) prevents the impressive Pcdo-1::GFP induction observed in the egl-9(-) and rhy-1(-) mutant animals. We have now mentioned this possible interpretation in the Results section. Furthermore, we like the idea of using genetic tricks to promote cysteine accumulation within C. elegans cells and tissues and will consider these approaches in future studies.

> *Issue 5: Second, there remain several major questions regarding the interpretation of the cysteine homeostasis pathway. How much specificity is involved for the RHY-1/CYSL-1/EGL-9/HIF-1 pathway to control cysteine homeostasis? Is the pathway able to sense cysteine directly or indirectly through its metabolites or redox status in general? Given the very low and high physiological concentrations of intracellular cysteine and glutathione (GSH, a major reserve for cysteine), respectively, there is a surprising lack of mention and testing of GSH metabolism.*

Response 5: Future studies are required to determine the specificity of the RHY-1/CYSL-1/EGL-9/HIF-1 pathway for the control of cysteine homeostasis. Our proposed mechanism, that H2S activates the HIF-1 pathway is based largely on the work of the Horvitz lab (Ma et al. 2012). They demonstrate that H2S promotes a direct inhibitory interaction between CYSL-1 and EGL-9, leading to activation of HIF-1. These findings align nicely with our genetic and pharmacological data. However, our work does not provide direct evidence as to the cysteine-derived metabolite that activates HIF-1. We propose H2S as a likely candidate.

We have added a note to the introduction regarding the role of GSH as a reservoir of excess cysteine and agree that future studies might find interesting links between CDO-1, GSH metabolism, and HIF-1.

> *Issue 6: In addition, what are the major similarities and differences of cysteine homeostasis pathways between C. elegans and other systems (HIF dependency, transcription vs post-transcriptional control)? These questions could be better discussed and noted with novel findings of the current study that are likely C. elegans specific or broadly conserved.*

Response 6: We have included a new section in the Discussion highlighting the nature of mammalian CDO1 regulation. We propose the hypothesis that a homologous pathway to the C. elegans RHY-1/CYSL-1/EGL9/HIF-1 pathway might operate in mammalian cells to sense high cysteine and induce CDO1 transcription. Importantly, all proteins in the C. elegans pathway have homologous counterparts in mammals. However, this hypothesis remains to be tested in mammalian systems.

> ***Reviewer #3 (Public Review):***
>
> *Major weaknesses of the paper include:*

> *Issue 7: the over-reliance on genetic approaches.*

Response 7: This is a fair critique. Our expertise is genetics. Our philosophy, which the reviewers may not share, is that there is no such thing as too much genetics!

> *Issue 8: the lack of novelty regarding prolyl hydroxylase-independent activities of EGL-9.*

Response 8: We believe the primary novelty of our work is establishing the intersection between the H2Ssensing HIF-1 pathway and cysteine catabolism governed by cysteine dioxygenase. Our demonstration that cdo-1 regulation operates largely independent of VHL-1 and EGL-9 prolyl hydroxylation is a mechanistic detail of this regulation and not the critical new finding. Although, we believe it does suggest where pathway analyses should be directed in the future. We also believe that our homeostatic feedback model for the regulation of HIF-1 (and cdo-1) by cysteine-derived H2S is new and exciting and provides insight into the logic of why HIF-1 might respond to H2S and promote the activity of cdo-1. Our work suggests that one reason for this intersection of hif-1 and cdo-1 is to sense and maintain cysteine homeostasis when cysteine is in excess.

> *Issue 9: the lack of biochemical approaches to probe the underlying mechanism of the prolyl hydroxylaseindependent activity of EGL-9.*

Response 9: While not the primary focus of our current manuscript, we agree that this is an exciting area of future research. To uncover the prolyl hydroxylase-independent activity of EGL-9, we agree that a combination of approaches will be required including, biochemical, structure-function, and genetic.

> *Major Issues We Feel the Authors Should Address:*
>
> *Issue 10: One particularly glaring concern is that the authors really do not know the extent to which the prolyl hydroxylase activity is (or is not) impacted by the H487A mutation in egl-9(rae276). If there is a fair amount of enzymatic activity left in this mutant, then it complicates interpretation. The paper would be strengthened if the authors could show that the egl-9(rae276) eliminates most if not all prolyl hydroxylase activity. In addition, the authors may want to consider doing RNAi for egl-9 in the egl-9(rae276) mutant as a control, as this would support the claim that whatever non-hydroxylase activity EGL-9 may have is indeed the causative agent for the elevation of CDO-1::GFP. Without such experiments, readers are left with the nagging concern that this allele is simply a hypomorph for the single biochemical activity of EGL-9 (i.e., the prolyl hydroxylase activity) rather than the more interesting, hypothesized scenario that EGL-9 has multiple biochemical activities, only one of which is the prolyl hydroxylase activity.*

Response 10: We have two lines of evidence that suggest the egl-9(rae276)-encoded H487A variant eliminates prolyl hydroxylase activity. First, Pan et al. 2007 (reference 57) demonstrate that when the equivalent histidine (H313) is mutated in human protein, that protein lacks detectible prolyl hydroxylase activity. Second, the phenotypic similarities caused by egl-9(rae276) and the vhl-1 null allele, ok161. Both alleles cause nearly identical activation of the Pcdo-1::GFP reporter transgene (Fig. 5C,D), and similarly impact the growth of the suox-1(gk738847) hypomorphic mutant (Table 1). This phenotypic overlap is highly relevant as the established role of VHL-1 is to recognize the hydroxyl mark conferred by the EGL-9 prolyl hydroxylase domain and promote the degradation of HIF-1. If EGL-9[H487A] had residual prolyl hydroxylase activity, we would expect the vhl-1(-) null mutant C. elegans to display more dramatic phenotypes than their egl-9(rae276) counterparts. This is not the case.

> *Issue 11: The authors observed that EGL-9 can inhibit HIF-1 and the expression of the HIF-1 target cdo-1 through a combination of activities that are (1) dependent on its prolyl hydroxylase activity (and subsequent VHL-1 activity that acts on the resulting hydroxylated prolines on HIF-1), and (2) independent of that activity. This is not a novel finding, as the authors themselves carefully note in their Discussion section, as this odd phenomenon has been observed for many HIF-1 target genes in multiple publications. While this manuscript adds to the description of this phenomenon, it does not really probe the underlying mechanism or shed light on how EGL-9 has these dual activities. This limits the overall impact and novelty of the paper.*

Response 11: See response to Issues #8.

> *Issue 12: Cysteine dioxygenases like CDO-1 operate in an oxygen-dependent manner to generate sulfites from cysteine. CDO-1 activity is dependent upon availability of molecular oxygen; this is an unexpected characteristic of a HIF-1 target, as its very activation is dependent on low molecular oxygen. Authors neither address this in the text nor experimentally, and it seems a glaring omission.*

Response 12: We agree this is an important point to raise within our manuscript. Although, despite its induction by HIF-1, there is no evidence that cdo-1 transcription is induced by hypoxia. In fact, in a genome wide transcriptomic study, cdo-1 was not found to be induced by hypoxia in C. elegans (Shen et al. 2005, reference 71).

We have newly commented on the use of molecular oxygen as a substrate by both EGL-9 and CDO-1 in our Discussion section. The mammalian oxygen-sensing prolyl hydroxylase (EGLN1) has been demonstrated to have high a Km value for O2 (high μM range). This likely allows EGLN1 to be poised to respond to small decreases in cellular oxygen from normal oxygen tensions. Clearly, CDO-1 also requires oxygen as a substrate, however the Km of CDO-1 for O2 is likely to be much lower, preventing sensitivity of the cysteine catabolism to physiological decreases in O2 availability. Although, to our knowledge, the CDO1 Km value for O2 has not been experimentally determined. We have added a new Discussion section where we address the conundrum about low oxygen inducing HIF-1 but oxygen being needed by CDO-1/CDO1.

> *Issue 13: The authors determined that the hypodermis is the site of the most prominent CDO-1::GFP expression, relevant to Figure 4. This claim would be strengthened if a negative control tissue, in the animal with the knockin allele, were shown. The hypodermal specific expression is a highlight of this paper, so it would make this article even stronger if they could further substantiate this claim.*

Response 13: Our claim that the hypodermis is the critical site of cdo-1 function is based on; i) our hands on experience looking at Pcdo-1::GFP, Pcdo-1::CDO-1::GFP, CDO-1::GFP (encoded by cdo-1(rae273)) and our reporting of these expression patterns in multiple figures throughout the manuscript and ii) the functional rescue of cdo-1(-) phenotypes by a cdo-1 rescue construct expressed by a hypodermal-specific promoter (col10). We agree that providing negative control tissues would modestly improve the manuscript. However, we do not think that adding these controls will substantially alter the conclusions of the paper. Importantly, we acknowledge this limitation of our work with the sentence, "However, we cannot exclude the possibility that CDO-1 also acts in other cells and tissues as well."

> *Minor issues to note:*
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> *Issue 14: Mutants for hif-1 and cysl-1 are sensitive to exogenous cysteine levels, yet loss of CDO-1 expression is not sufficient to explain this phenomenon, suggesting other targets of HIF-1 are involved. Given the findings the authors (and others) have had*

> *showing a role for RHY-1 in sulfur amino acid metabolism, shouldn't the authors consider testing rhy-1 mutants for sensitivity to exogenous cysteine?*

Response 14: To test the hypothesis that rhy-1(-) C. elegans might be sensitive to supplemental cysteine, we cultured wild type and rhy-1(-) animals on 0, 100, and 1000µM supplemental cysteine. At 0 and 100µM supplemental cysteine, neither wild-type nor rhy-1(-) animals display any lethality suggesting rhy-1 is not required for survival in the face of excess cysteine (Fig. 3D,E). We also cultured these same strains on 1000µM supplemental cysteine, a concentration that is highly toxic to wild-type animals (100% lethality). rhy1(-) animals were resistant to 1000µM supplemental cysteine with a substantial fraction of the population surviving overnight exposure to this lethal dose of cysteine. Similarly, egl-9(-) mutant C. elegans were also resistant to 1000µM supplemental cysteine. We propose that loss of egl-9 or rhy-1 activates HIF-1-mediated transcription which is priming these mutants to cope with the lethal dose of cysteine. These data are now presented in Figure 3D-F and presented in the Results section.

> *Issue 15: The cysteine exposure assay was performed by incubating nematodes overnight in liquid M9 media containing OP50 culture. The liquid culture approach adds two complications: (1) the worms are arguably starving or at least undernourished compared to animals grown on NGM plates, and (2) the worms are probably mildly hypoxic in the liquid cultures, which complicates the interpretation.*

Response 15: We agree that it is possible that animals growing overnight in liquid culture are undernourished and mildly hypoxic. However, we are confident in our data interpretation as all our experiments are appropriately controlled. Meaning, control and experimental groups were all grown under the same liquid culture conditions. Thus, these animals would all experience the same stressors that come with liquid culture. Importantly, we never make comparisons between groups that were grown under different culture conditions (i.e. solid media vs. liquid culture).

> *Issue 16: An easily addressable concern is the wording of one of the main conclusions: that cdo-1 transcription is independent of the canonical prolyl hydroxylase function of EGL-9 and is instead dependent on one of EGL-9's non-canonical, non-characterized functions. There are several points in which the wording suggests that CDO-1 toxicity is independent of EGL-9. In their defense, the authors try to avoid this by saying, "EGL-9 PHD," to indicate that it is the prolyl hydroxylase function of EGL-9 that is not required for CDO-1 toxicity. However, this becomes confusing because much of the field uses PHD and EGL-9/EGLN as interchangeable protein names. The authors need to be clear about when they are describing the prolyl hydroxylase activity of EGL-9 rather than other (hypothesized) activities of EGL-9 that are independent of the prolyl hydroxylase activity.*

Response 16: We appreciate the reviewer alerting us to this practice within the field. To avoid confusion, we have removed the “PHD” abbreviation from our manuscript and explicitly referred to the “prolyl hydroxylase domain” where relevant.

> *Issue 17: The authors state in the text, "the egl-9; suox-1 double mutants are extremely sick and slow growing." We appreciate that their "health" assay, based on the exhaustion of food from the plate, is qualitative. We also appreciate that it is a functional measure of many factors that contribute to how fast a population of worms can grow, reproduce, and consume that lawn of food. However, unless they do a lifespan assay and/or measure developmental timing and specifically determine that the double mutant animals themselves are developing and/or growing more slowly, we do not think it is appropriate to use the words "slow growing" to describe the population. As they point out, the rate of consumption of food on the plate in their health assay is determined by a*

> *multitude and indeed a confluence of factors; the growth rate is one specific one that is commonly measured and has an established meaning.*

Response 17: We see how the phrase ‘slow growing’ might imply a phenotype that we have not actually assessed with this assay. Therefore, we have removed all claims about “slow growth” of the strains presented in Table 1 and have highlighted the assay more overtly in the results section. For example; “While egl-9(-) and suox-1(gk738847) single mutant animals are healthy under standard culture conditions, the egl-9(-); suox1(gk738847) double mutant animals are extremely sick and require significantly more days to exhaust their E. coli food source under standard culture conditions (Table 1).”

> ***Reviewer #1 (Recommendations For The Authors):***
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> *Issue 18: Relevance could be addressed further in the text.*

Response 18: We have added additional context for our work in the Discussion section. Please see our response to Issues #5, 6, 12, and 24.

> *Issue 19: Better appreciation and integration of the manuscript's findings with published studies would be appropriate.*

Response 19: We have added additional context for our work in the Discussion section. Please see our response to Issues #5, 6, 12, and 24.

> *Issue 20: It might be perhaps relevant to test whether cdo-1 is relevant for hypoxia resistance since it appears to be a key target for hif-1.*

Response 20: We agree that this is an interesting future direction, however given that cdo-1 mRNA is not induced by hypoxia (Shen et al. 2005) we have not prioritized these experiments for the current manuscript.

> *Issue 21: "egl-9 inhibits cdo-1 transcription in a prolyl-hydroxylase and VHL-1-independent manner" should be tempered. vhl-1 mutants and egl-9 hydroxylase point mutant still have significant induction of the reporter.*

Response 21: Thank you for identifying this oversight. We have modified the Figure 5 legend title to read, “egl9 inhibits cdo-1 transcription in a largely prolyl-hydroxylase and VHL-1-independent manner.”

> *Issue 22: Please use line numbers in the future for easier tracking of comments.*

Response 22: We shall.

> *Issue 23: Abstract and elsewhere, "high cysteine activates...", should be rephrased to "high levels of cysteine".*

Response 23: We have made this change throughout the manuscript.

> ***Reviewer #3 (Recommendations For The Authors):***
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> *Issue 24: The authors discuss CDO1 in the context of tumorigenesis, as well as the potential regulation between cysteine and the hypoxia response pathway. Thus, I was surprised that there was no mention of the foundational Bill Kaelin paper (Briggs et al 2016) showing how the accumulation of cysteine is related to tumorigenesis, and that cysteine is a direct activator of EglN1. Puzzling that CDO1 is a tumor suppressor: you lose*

> *it, cysteine can accumulate and activate EglN1, causing HIF1 turnover. How do the authors reconcile their results with this paper? I was also surprised that there was no mention in the Discussion of the role of hydrogen sulfide, cysteine metabolism, and CTH and CBS in oxygen sensation in the carotid body given the role they play there. Seems important to discuss this issue.*

Response 24: We have added new sections to our Discussion that consider the relationship between our work and Briggs et al. 2016 as well as mentioned the role of CTH and H2S in the mammalian carotid body.

> *Issue 25: The abstract has a variety of contradictory statements. For example, the authors state that "HIF-1mediated induction of cdo-1 functions largely independent of EGL-9," but then go on to conclude in the final sentence that cysteine stimulates H2S production, which then activates EGL-9 signaling, which then increases HIF-1-mediated transcription of cdo-1. A quick reading of the abstract leaves the reader uncertain whether EGL-9 is or is not involved in this regulation of cdo-1 expression. In addition, the conclusion sentence implies that activation of the EGL-9 pathway increases HIF-1-mediated transcription, yet it is well established that EGL-9 is an inhibitor of HIF-1. The abstract fails to deliver a clear summary of the paper's conclusions. Perhaps consider this alternative (changes in capital letters):*
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> *The amino acid cysteine is critical for many aspects of life, yet excess cysteine is toxic. Therefore, animals require pathways to maintain cysteine homeostasis. In mammals, high cysteine activates cysteine dioxygenase, a key enzyme in cysteine catabolism. The mechanism by which cysteine dioxygenase is regulated remains largely unknown. We discovered that C. elegans cysteine dioxygenase (cdo-1) is transcriptionally activated by high cysteine and the hypoxia inducible transcription factor (hif-1). hif-1- dependent activation of cdo-1 occurs downstream of an H2S-sensing pathway that includes rhy-1, cysl-1, and egl-9. cdo-1 transcription is primarily activated in the hypodermis where it is sufficient to drive sulfur amino acid metabolism. EGL-9 and HIF-1 are core members of the cellular hypoxia response. However, we demonstrate that the mechanism of HIF-1-mediated induction of cdo-1 IS largely independent of EGL-9 prolyl hydroxylASE ACTIVITY and the von Hippel-Lindau E3 ubiquitin ligase. We propose that the REGULATION OF cdo-1 BY HIF-1 reveals a negative feedback loop for maintaining cysteine homeostasis. High cysteine stimulates the production of an H2S signal. H2S then ACTS THROUGH the rhy-1/cysl-1/egl-9 signaling pathway DISTINCTLY FROM THEIR ROLE IN HYPOXIA RESPONSE TO INCREASE HIF-1-mediated transcription of cdo-1, promoting degradation of cysteine via CDO-1.*

Response 25: We agree that the abstract could be clearer. We believe this concern stems from the fact that we did not discuss our initial screen in the abstract. Thus, we failed to establish a role for egl-9 in the regulation of cdo-1. To remedy this, we have modified the abstract as suggested by the reviewer and added additional context. We believe that these changes improve the clarity of the Abstract substantially.

> *Issue 26: An easily addressable concern involves the "dark" microscopy controls showing lack of fluorescence from a nematode. In these dark negative control micrographs, the authors should draw dotted outlines around where the worms are or include a brightfield image next to the fluorescence image. On a computer screen, it is in fact possible to make out the worms. Yet, when printed out, the reader must assume there are worms in the dark images. Additionally, we realize that adjusting fluorescence so that wild-type CDO-1 expression can be seen will result in oversaturation of the egl-9 and rhy-1; cdo-1 doubles; however, this would be a useful figure to add into the supplement to both provide a normal reference of CDO-1 low-level expression and a demonstration*

> *of just how bright it is in the mutant backgrounds. It would also be useful for you to please report your exposure settings for purposes of reproducibility.*

Response 26: As suggested, we have added dotted lines around the location of the C. elegans animals in all images where GFP expression is low or basal. We have also reported the exposure times for each image in the appropriate figure legends.

> *Issue 27: This title is quite generic and doesn't even mention the main players (CDO-1 and sulfite metabolism).*

Response 27: We have updated our title to call attention to cysteine dioxygenase. The improved title is: "Hypoxia-inducible factor induces cysteine dioxygenase and promotes cysteine homeostasis in Caenorhabditis elegans"

> *Issue 28: The authors mention two disorders in which CDO-1 plays a pathogenic role: MoCD and ISOD. We recommend switching the order in which the authors mention these, as the remainder of the paragraph is about MoCD. Also, they should write out the number "2" in the first sentence of that paragraph.*

Response 28: We have made the suggested changes.

> *Issue 29: The authors state in the main text, "...to ubiquitinate HIF-1, targeting it for degradation by the proteosome." Here, they should refer to the pathway in Figure 5a.*

Response 29: We have made the suggested change.

> *Issue 30: The authors state in the main text, "Elements of the HIF-1 pathway have emerged..." which is vague and confusingly worded. Change to, "Members of the HIF-1 pathway and its targets have emerged from C. elegans genetic studies."*

Response 30: We have made the suggested change.

> *Issue 31: Clarify in the figure legends that supplemental cysteine did not affect the mortality of worms that were imaged.*

Response 31: We have added this note to Figure 3A and Figure S3A.

> *Issue 32: Figure 1b. "the cdo-1 promoter is shown..." Add: "as a straight line" to the end of this phrase.*

Response 32: We have made the suggested change.

> *Issue 33: The authors should consider changing the red text in Figure 1 to magenta, which tends to be more readable for people who have limited color vision.*

Response 33: We have adjusted the colors in Figure 1 as suggested.

> *Issue 34: Figure 2, legend title. Consider changing "hif-1" to "HIF-1," as well as rhy-1, cysl-1, and egl-9. In this case, they are talking about proteins, not mutants or genes. This will make the paper easier to follow for readers who lack a C. elegans background.*

Response 34: We have made the suggested change.

> *Issue 35: Figure 5, caption text. "...indicates weak similarity." Add, "amongst species compared."*

Response 35: We have made the suggested change.

> *Issue 36: It is starting to become a standard for showing the datapoints in bar graphs. Although this is done in many graphs in the paper, it should also be done for Figure S1 and Figure 4C.*

Response 36: We have made the suggested change.

> *Issue 37: An extensive ChIP-seq and RNA-seq analysis of C. elegans HIF-1 was recently published (Vora et al, 2022), which the authors should reference in support of the regulation of CDO-1 transcription by HIF-1 in their description of published expression studies of the pathway (Results section, page 4). Indeed, Vora et al were key generators of the ChIP-seq data cited in Warnhoff et al but not included as authors in the ModERN/ModENCODE publication: their contributions were published separately in Vora et al and should be acknowledged equivalently.*

Response 37: We appreciate the reviewer pointing this detail out and we have added the correct citation as indicated.